

BaseCore of the Rigid Identity Framework

Autonomous Mathematical Source Text for Foundational Dynamics, Identity Rigidity,
Regime Constraints, and Operational-Criterion Grammar

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Abstract

This document is the BaseCore of the Rigid Identity Framework: the autonomous mathematical base layer of the source tree. It selectively normalizes formal material from Papers 1--6 only where theorem ownership belongs in the base layer. Papers 7--9 remain downstream and are not required for the validity of any theorem stated here. BaseCore proves contractive projection dynamics, typed computable witnesses, parameterwise non-collapse under explicit anti-constant assumptions, inverse-limit identity under non-finite-determination assumptions, conditional rigidity under admissible perturbation models, conditional non-simulability of CCR targets by finite simulator classes, operational-criterion grammar, and falsation / claim-boundary ledgers. It does not prove phenomenality, metaphysical subjectivity, human equivalence, runtime instantiation, bridge admissibility, or external adjudication. The LaTeX tree is the source of truth; the PDF is a compiled artifact only.

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Part I

Foundational Dynamics

1 Scope, System Boundary, and Non-Inference Note

What BaseCore does. This document proves the mathematical base results that later layers may use: projection existence and uniqueness, strict contraction, unique fixed points, compact attractor families, parameterwise non-collapse under explicit anti-constant assumptions, inverse-limit identity under non-finite-determination assumptions, conditional rigidity, conditional non-simulability, and operational-criterion grammar.

What BaseCore is not. BaseCore is not the whole QICN corpus. It is not a bridge program, not runtime validation, not a phenomenality theory, not a human-equivalence claim, and not an external-adjudication layer.

What should not be inferred. Nothing in BaseCore alone licenses claims about consciousness, subjectivity, phenomenality, moral status, human-machine equivalence, bridge admissibility, or deployment readiness. Those are downstream burdens, not consequences of the theorems below.

Source-of-truth rule. The LaTeX tree is the source of truth. Compiled PDFs, release bundles, and runtime artifacts are derived surfaces only.

2 Introduction

2.1 Purpose

BaseCore is the canonical base layer of the Rigid Identity Framework. Its role is narrower and stronger than a broad architectural survey: it owns only the theorem-level mathematics that downstream papers may depend on without making BaseCore depend on them.

2.2 Methodology

We use standard tools from:

- Hilbert-space geometry and metric projections;
- contraction mappings and fixed-point theory;
- compactness and continuity arguments;
- projective systems and inverse limits;
- assumption-led claim-boundary discipline.

2.3 Scope discipline

This section is pure mathematics. The symbols “identity,” “regime,” and “criterion” are structural labels only. Their ordinary-language interpretations are not part of any proof unless explicitly formalized as assumptions.

3 Minimal Base Hypotheses

Hypothesis 3.1 (H1: Projection Structure). Let $\mathcal{H}$ be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. Let $\mathcal{I} \subset \mathcal{H}$ be non-empty, closed, and convex.

Hypothesis 3.2 (H2: Contraction Operator). Let $K : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded linear operator with operator norm $\|K\| < 1$.

Hypothesis 3.3 (H3: Completeness). The ambient space $\mathcal{H}$ is complete.

Hypothesis 3.4 (H4: Parameter Compactness). Let $\mathcal{U}$ be a compact metric space and let $\Gamma : \mathcal{U} \rightarrow \mathcal{H}$ be uniformly continuous.

Proposition 3.5 (Base sufficiency of H1--H4). *Hypotheses H1--H4 are sufficient for:*

1. *existence and uniqueness of the metric projection onto $\mathcal{I}$;*
2. *strict contraction of the transition operators T_u ;*
3. *existence and uniqueness of the fixed point f_u^* for each $u \in \mathcal{U}$;*
4. *compactness of the attractor family $\mathcal{A}^* = \{f_u^* : u \in \mathcal{U}\}$.*

They do not, by themselves, imply parameterwise non-collapse.

Proof. The four items are exactly the content of Theorems 4.2, 5.2, 6.1, and 6.3. Parameterwise non-collapse requires an additional anti-constant hypothesis introduced in Section 7. $\square$

4 The Projection Operator

Definition 4.1 (Metric Projection). For $\Psi \in \mathcal{H}$, define the metric projection onto $\mathcal{I}$ by

$$\mathfrak{N}(\Psi) := \arg \min_{\Phi \in \mathcal{I}} \|\Psi - \Phi\|.$$

Theorem 4.2 (Existence and Uniqueness of Projection). *Under Hypothesis 3.1, the projection $\mathfrak{N}(\Psi)$ exists and is unique for every $\Psi \in \mathcal{H}$.*

Proof. Existence follows because a minimizing sequence in the closed convex set $\mathcal{I}$ is Cauchy by the parallelogram law, hence converges in the complete space $\mathcal{H}$ to a point of $\mathcal{I}$ that attains the infimum. Uniqueness follows from strict convexity of the norm induced by the Hilbert inner product: if two minimizers existed, their midpoint would produce the same minimum and the parallelogram identity would force the two minimizers to coincide. $\square$

Lemma 4.3 (Non-Expansiveness of the Projection). *The map $\mathfrak{N} : \mathcal{H} \rightarrow \mathcal{I}$ is non-expansive:*

$$\|\mathfrak{N}(\Psi_1) - \mathfrak{N}(\Psi_2)\| \leq \|\Psi_1 - \Psi_2\| \quad \forall \Psi_1, \Psi_2 \in \mathcal{H}.$$

Proof. This is the standard Hilbert-space projection inequality. Let $\Phi_i = \mathfrak{N}(\Psi_i)$. The projection characterization

$$\langle \Psi_i - \Phi_i, \Phi - \Phi_i \rangle \leq 0 \quad \forall \Phi \in \mathcal{I}$$

applied with $\Phi = \Phi_2$ and $\Phi = \Phi_1$ yields

$$\|\Phi_1 - \Phi_2\|^2 \leq \langle \Psi_1 - \Psi_2, \Phi_1 - \Phi_2 \rangle \leq \|\Psi_1 - \Psi_2\| \|\Phi_1 - \Phi_2\|.$$

If $\Phi_1 \neq \Phi_2$, divide by $\|\Phi_1 - \Phi_2\|$. $\square$

5 The Transition Operator

Definition 5.1 (Transition Operator). For each $u \in \mathcal{U}$, define

$$T_u(\Psi) := \aleph(K\Psi + \Gamma(u)).$$

Theorem 5.2 (Contractivity). *Under Hypotheses 3.1 and 3.2, each T_u is a strict contraction with Lipschitz constant at most $\|K\|$:*

$$\|T_u(\Psi_1) - T_u(\Psi_2)\| \leq \|K\| \|\Psi_1 - \Psi_2\|.$$

Proof. By Lemma 4.3,

$$\begin{aligned} \|T_u(\Psi_1) - T_u(\Psi_2)\| &= \|\aleph(K\Psi_1 + \Gamma(u)) - \aleph(K\Psi_2 + \Gamma(u))\| \\ &\leq \|K(\Psi_1 - \Psi_2)\| \leq \|K\| \|\Psi_1 - \Psi_2\|. \end{aligned}$$

Since $\|K\| < 1$, strict contraction follows. $\square$

6 Existence and Compactness of Attractors

Theorem 6.1 (Unique Fixed Point). *Under Hypotheses 3.1--3.3, for each $u \in \mathcal{U}$ there exists a unique $f_u^* \in \mathcal{H}$ such that*

$$T_u(f_u^*) = f_u^*.$$

Moreover, for every $\Psi_0 \in \mathcal{H}$,

$$\lim_{n \rightarrow \infty} T_u^n(\Psi_0) = f_u^*.$$

Proof. By Theorem 5.2, each T_u is a strict contraction on the complete metric space $(\mathcal{H}, \|\cdot\|)$. The Banach fixed-point theorem yields existence, uniqueness, and convergence of iterates. $\square$

Definition 6.2 (Attractor Family). Define the attractor family

$$\mathcal{A}^* := \{f_u^* : u \in \mathcal{U}\}.$$

Theorem 6.3 (Compactness of the Attractor Family). *Under Hypotheses 3.1--3.4, the set $\mathcal{A}^*$ is compact in $\mathcal{H}$.*

Proof. Let $F : \mathcal{U} \rightarrow \mathcal{H}$ be given by $F(u) = f_u^*$. The fixed points depend continuously on u because the contraction constant is uniform and Γ is uniformly continuous on the compact parameter space. Since $\mathcal{U}$ is compact and $\mathcal{A}^* = F(\mathcal{U})$, the image is compact. $\square$

7 Parameterwise Non-Collapsibility

Definition 7.1 (Constant Subspace). Whenever the ambient realization distinguishes constant elements, let $\mathcal{N} \subset \mathcal{H}$ denote the closed subspace of constants.

Definition 7.2 (Parameterwise Non-Collapsible Family). A family $\{T_u\}_{u \in \mathcal{U}}$ is parameterwise non-collapsible if

$$f_u^* \notin \mathcal{N} \quad \forall u \in \mathcal{U}.$$

Hypothesis 7.3 (H5: No Constant Fixed Points). For every parameter $u \in \mathcal{U}$ and every constant element $c \in \mathcal{N}$,

$$T_u(c) \neq c.$$

Theorem 7.4 (Structural Non-Collapsibility). *Under Hypotheses 3.1--7.3, the family is parameterwise non-collapsible:*

$$f_u^* \notin \mathcal{N} \quad \forall u \in \mathcal{U}.$$

Proof. Fix $u \in \mathcal{U}$. Suppose by contradiction that $f_u^* \in \mathcal{N}$. Write $f_u^* = c$ with $c \in \mathcal{N}$. Since f_u^* is a fixed point,

$$T_u(c) = T_u(f_u^*) = f_u^* = c,$$

contradicting Hypothesis 7.3. Therefore $f_u^* \notin \mathcal{N}$. $\square$

Remark 7.5 (What H5 does not say). Hypothesis 7.3 is parameterwise. It does not merely say that *some* parameter breaks constant collapse. It excludes constant fixed points for *every* parameter. This is exactly the quantifier strength needed for Theorem 7.4.

8 Summary of Base Results

1. H1--H4 give a complete theorem-tight base for projection existence, strict contraction, unique fixed points, and compact attractor families.
2. H5 is a separate anti-constant assumption block used only for parameterwise non-collapse.
3. No empirical, phenomenological, biological, or runtime claim is inferred at this level.
4. Later sections may add conditional identity, rigidity, and criterion theorems, but only by introducing their own explicit assumption blocks.

Part II

Typed Computable Model and State Spectral Gap

9 Typed Computable Model: Embedded Constant-Disk Witness

This section repairs the type leaks of the previous computable model. The purpose is narrow: provide a fully typed witness for the transition grammar and the state spectral gap. It does *not* by itself certify parameterwise non-collapse; that burden is handled separately below.

9.1 Golden Mean Parameter Space

Definition 9.1 (Golden Mean Shift Parameter Space). Let $\Sigma = \{0, 1\}$ and

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The Golden Mean Shift $S_A \subset \Sigma^{\mathbb{N}}$ is the subshift of finite type consisting of all binary sequences that avoid the block 11.

Definition 9.2 (Parameter Encoding). Define

$$u(x) = \sum_{i=1}^{\infty} x_i \phi^{-i}, \quad \phi = \frac{1 + \sqrt{5}}{2}.$$

The image $\mathcal{U} := u(S_A) \subset [0, 1]$ is compact.

9.2 Typed Banach Witness

Definition 9.3 (Ambient Space and Constant Embedding). Let

$$X = [0, 1], \quad E = \mathbb{R}^2, \quad \mathcal{B} = C(X, E)$$

with norm

$$\|f\|_\infty := \sup_{s \in X} \|f(s)\|_2.$$

Define the constant embedding

$$j : E \rightarrow \mathcal{B}, \quad j(v)(s) = v.$$

Definition 9.4 (Invariant Constant Disk). Let

$$D := \{v \in E : \|v\|_2 \leq 1\}, \quad \mathcal{I}_c := j(D) \subset \mathcal{B}.$$

Definition 9.5 (Averaging Operator). Define

$$A : \mathcal{B} \rightarrow E, \quad A(f) := \int_0^1 f(\tau) d\tau.$$

Definition 9.6 (Disk Projection). Define the Euclidean projection

$$P_E : E \rightarrow D, \quad P_E(v) = \begin{cases} v, & \|v\|_2 \leq 1, \\ \frac{v}{\|v\|_2}, & \|v\|_2 > 1. \end{cases}$$

Definition 9.7 (Typed Projection onto the Constant Disk). Define

$$\mathfrak{N}_{\mathcal{B}} : \mathcal{B} \rightarrow \mathcal{I}_c, \quad \mathfrak{N}_{\mathcal{B}}(f) := j(P_E(A(f))).$$

Definition 9.8 (Constant-Valued Contraction Operator). Define

$$\widetilde{K} : \mathcal{B} \rightarrow \mathcal{B}, \quad (\widetilde{K}f)(s) := \frac{1}{2} \int_0^1 f(\tau) d\tau.$$

Definition 9.9 (Lifted Input Coupling). Let $\Gamma : \mathcal{U} \rightarrow E$ be

$$\Gamma(u) = (u, 0).$$

Define its constant lift

$$\Gamma_{\mathcal{B}} : \mathcal{U} \rightarrow \mathcal{B}, \quad \Gamma_{\mathcal{B}}(u) := j(\Gamma(u)).$$

Definition 9.10 (Typed Transition Operator). For each $u \in \mathcal{U}$, define

$$T_u : \mathcal{B} \rightarrow \mathcal{I}_c, \quad T_u(f) := \mathfrak{N}_{\mathcal{B}}(\widetilde{K}f + \Gamma_{\mathcal{B}}(u)).$$

Remark 9.11 (Typing discipline). No unembedded element of E is treated as an element of $\mathcal{B}$. The embedding j is always written explicitly. This removes the earlier type leak in which a disk in E was silently treated as a subset of a function space.

9.3 Basic Operator Properties

Lemma 9.12 (The Embedding is Isometric). *For every $v \in E$,*

$$\|j(v)\|_\infty = \|v\|_2.$$

Proof. By definition,

$$\|j(v)\|_\infty = \sup_{s \in X} \|j(v)(s)\|_2 = \sup_{s \in X} \|v\|_2 = \|v\|_2.$$

□

Lemma 9.13 (The Disk Projection is Non-Expansive). *For all $v, w \in E$,*

$$\|P_E(v) - P_E(w)\|_2 \leq \|v - w\|_2.$$

Proof. P_E is the metric projection onto the closed convex set $D \subset \mathbb{R}^2$, so the standard Hilbert-space non-expansiveness property applies. □

Lemma 9.14 (The Typed Projection is Non-Expansive). *For all $f, g \in \mathcal{B}$,*

$$\|\aleph_{\mathcal{B}}(f) - \aleph_{\mathcal{B}}(g)\|_\infty \leq \|f - g\|_\infty.$$

Proof. Using the isometry of j , the non-expansiveness of P_E , and Jensen's inequality for the averaging operator,

$$\begin{aligned} \|\aleph_{\mathcal{B}}(f) - \aleph_{\mathcal{B}}(g)\|_\infty &= \|P_E(A(f)) - P_E(A(g))\|_2 \\ &\leq \|A(f) - A(g)\|_2 \\ &= \left\| \int_0^1 (f(\tau) - g(\tau)) d\tau \right\|_2 \\ &\leq \int_0^1 \|f(\tau) - g(\tau)\|_2 d\tau \leq \|f - g\|_\infty. \end{aligned}$$

□

Lemma 9.15 (Norm of the Lifted Contraction).

$$\|\widetilde{K}\| \leq \frac{1}{2}.$$

Proof. For any $f \in \mathcal{B}$,

$$\|\widetilde{K}f\|_\infty = \left\| \frac{1}{2} \int_0^1 f(\tau) d\tau \right\|_2 \leq \frac{1}{2} \int_0^1 \|f(\tau)\|_2 d\tau \leq \frac{1}{2} \|f\|_\infty.$$

□

Theorem 9.16 (Typed Contraction on the Computable Witness). *For each $u \in \mathcal{U}$, the map $T_u : \mathcal{B} \rightarrow \mathcal{I}_c$ is a strict contraction:*

$$\|T_u(f) - T_u(g)\|_\infty \leq \frac{1}{2} \|f - g\|_\infty.$$

Proof. Combine non-expansiveness of $\aleph_{\mathcal{B}}$ with the bound on $\widetilde{K}$:

$$\|T_u(f) - T_u(g)\|_\infty \leq \|\widetilde{K}f - \widetilde{K}g\|_\infty \leq \frac{1}{2} \|f - g\|_\infty.$$

□

Corollary 9.17 (Unique Fixed Point in the Typed Witness). *For each $u \in \mathcal{U}$, there exists a unique $f_u^* \in \mathcal{I}_c$ such that*

$$T_u(f_u^*) = f_u^*.$$

Proof. The Banach fixed-point theorem applies on the complete metric space $(\mathcal{B}, \|\cdot\|_\infty)$. $\square$

Proposition 9.18 (Typed Non-Vacuity of the Computable Model). *The embedded constant-disk Golden Mean model is a fully typed, non-vacuous witness of the transition grammar:*

1. *every operator has explicit domain and codomain;*
2. *the transition map T_u is well-defined on $\mathcal{B}$;*
3. *the family admits unique fixed points and a compact parameter image;*
4. *the state contraction bound is explicit and computable.*

Proof. Items (1)–(3) follow from the definitions above and the contraction theorem. Compactness of the parameter image follows because $\mathcal{U}$ is compact and $u \mapsto f_u^*$ is continuous under a uniform contraction constant. $\square$

Remark 9.19 (Why this witness does not certify H5). The image of T_u lies in the constant subspace $\mathcal{I}_c$. Therefore its fixed point is constant by construction. This witness is correct and useful for typing and state spectral analysis, but it cannot certify the parameterwise anti-constant hypothesis H5 from Section 7.

9.4 Separate Anti-Collapse Witness

Definition 9.20 (Affine Non-Constant Witness Space). Let

$$\mathcal{H}_{ac} := L^2([0, 1]), \quad \mathcal{N} := \{c\mathbf{1} : c \in \mathbb{R}\}, \quad h(s) := \sin(2\pi s).$$

Choose r with $0 < r < \|h\|_{L^2}$ and define the closed convex affine set

$$\mathcal{I}_{ac} := h + \{g \in \mathcal{H}_{ac} : \|g\|_{L^2} \leq r\}.$$

Lemma 9.21 (The Anti-Collapse Witness Set Avoids Constants).

$$\mathcal{I}_{ac} \cap \mathcal{N} = \emptyset.$$

Proof. The distance from h to the constant subspace $\mathcal{N}$ equals $\|h\|_{L^2}$ because h has mean zero and is orthogonal to constants. Since $r < \|h\|_{L^2}$, the radius- r ball around h does not intersect $\mathcal{N}$. $\square$

Definition 9.22 (Anti-Collapse Witness Dynamics). Let $\mathcal{U} = [0, 1]$, choose $\beta \in (0, 1/2)$, and define

$$K_{ac}f := \beta f, \quad \Gamma_{ac}(u) := \frac{u}{4}h.$$

Let P_{ac} denote the metric projection onto $\mathcal{I}_{ac}$, and define

$$T_u^{ac}(f) := P_{ac}(K_{ac}f + \Gamma_{ac}(u)).$$

Proposition 9.23 (Separate Witness for H5). *The family $\{T_u^{ac}\}_{u \in \mathcal{U}}$ satisfies Hypotheses 3.1–7.3 of Section 1.*

Proof. 1. $\mathcal{H}_{ac}$ is a Hilbert space and $\mathcal{I}_{ac}$ is non-empty, closed, and convex.

2. $\|K_{ac}\| = \beta < 1$.

3. Completeness is inherited from $L^2([0, 1])$.
4. $\mathcal{U} = [0, 1]$ is compact and Γ_{ac} is continuous.
5. For every constant $c \in \mathcal{N}$ and every $u \in \mathcal{U}$, the value $T_u^{\text{ac}}(c)$ belongs to $\mathcal{I}_{\text{ac}}$, while $\mathcal{I}_{\text{ac}} \cap \mathcal{N} = \emptyset$. Hence $T_u^{\text{ac}}(c) \neq c$.

□

Remark 9.24. BaseCore therefore distinguishes two roles cleanly: the Golden Mean witness certifies typed operator correctness and state contraction; the affine anti-collapse witness certifies the stronger H5 burden needed for parameterwise non-collapse.

10 State Spectral Gap and Parameter-Family Persistence

Theorem 10.1 (State Spectral Gap). *Fix $u \in \mathcal{U}$ in the typed computable model. Assume T_u is Fréchet differentiable at its fixed point f_u^* . Then*

$$\rho(DT_u(f_u^*)) \leq \|DT_u(f_u^*)\| \leq \|\widetilde{K}\| \leq \frac{1}{2}.$$

Consequently the state spectral gap satisfies

$$\Delta_{\text{state}} \geq 1 - \|\widetilde{K}\| \geq \frac{1}{2}.$$

Proof. At every point where the projection P_E is Fréchet differentiable, its derivative has operator norm at most 1 because P_E is non-expansive. Since $\aleph_{\mathcal{B}} = j \circ P_E \circ A$,

$$DT_u(f_u^*) = D\aleph_{\mathcal{B}}(\widetilde{K}f_u^* + \Gamma_{\mathcal{B}}(u)) \circ \widetilde{K},$$

and therefore

$$\|DT_u(f_u^*)\| \leq \|D\aleph_{\mathcal{B}}\| \|\widetilde{K}\| \leq \frac{1}{2}.$$

The spectral-radius bound follows from $\rho(L) \leq \|L\|$ for bounded linear operators. □

Theorem 10.2 (Relaxation-Time Bound). *Under the hypotheses of Theorem 10.1, any discrete-to-continuous relaxation convention of the form*

$$\tau_{\text{state}} = -\frac{1}{\ln \rho}$$

with $\rho = \rho(DT_u(f_u^))$ gives*

$$\tau_{\text{state}} \leq \frac{1}{\ln 2}.$$

Proof. Theorem 10.1 gives $\rho \leq 1/2$. Since $-\ln(\rho) \geq \ln 2$, the bound follows. □

Proposition 10.3 (Parameter-Family Persistence is not State Spectrum). *Let $F : \mathcal{U} \rightarrow \mathcal{B}$ be the fixed-point family $F(u) = f_u^*$. Any persistence of parameter labels or family coordinates belongs to the variation of F over $\mathcal{U}$, not to the state-transition spectrum of $DT_u(f_u^*)$ for fixed u .*

Proof. The state-transition spectrum is defined for the linearization of T_u at fixed parameter u . Family-level transport instead concerns the map F between parameter space and fixed-point set. These are different operators on different domains. Therefore a family-persistence mode cannot be used to introduce a unit eigenvalue into the contraction spectrum of $DT_u(f_u^*)$. □

Remark 10.4 (What has been removed). The previous text incorrectly mixed strict contraction with a state-spectrum eigenvalue $\lambda = 1$. BaseCore no longer does that. Any neutral or persistent mode must now be interpreted as a parameter-family bookkeeping mode, not as an eigenmode of the contractive state Jacobian.

Part III

Identity, Rigidity, and Conditional Non-Simulability

11 Projective Systems and Inverse-Limit Identity

11.1 Projective Setup

Definition 11.1 (Projective System of State Spaces). Let $\{S_t\}_{t \in \mathbb{N}}$ be a family of non-empty state spaces together with projection maps

$$\pi_{t+1 \rightarrow t} : S_{t+1} \rightarrow S_t$$

satisfying the compatibility rule

$$\pi_{t+1 \rightarrow t} \circ \pi_{t+2 \rightarrow t+1} = \pi_{t+2 \rightarrow t} \quad \forall t \in \mathbb{N}.$$

Definition 11.2 (Identity as Inverse Limit). The identity object associated with the projective system is

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_{t \in \mathbb{N}} \in \prod_{t \in \mathbb{N}} S_t \mid \pi_{t+1 \rightarrow t}(x_{t+1}) = x_t \ \forall t \right\}.$$

Remark 11.3 (Identity neutrality). In BaseCore, $\mathcal{I}$ is a structural inverse-limit object only. It carries no automatic phenomenological, biological, or metaphysical interpretation.

Hypothesis 11.4 (Topological Regularity). Each S_t is compact Hausdorff and each $\pi_{t+1 \rightarrow t}$ is continuous and surjective.

Hypothesis 11.5 (NFD: Non-Finite Determination). For every $t \in \mathbb{N}$, the canonical projection

$$p_t : \mathcal{I} \rightarrow S_t, \quad p_t((x_k)_k) = x_t$$

is not injective. Equivalently,

$$\forall t \ \exists x, y \in \mathcal{I} \quad x \neq y \quad \text{and} \quad p_t(x) = p_t(y).$$

Remark 11.6 (Persistent future branching as an operational form of NFD). A sufficient operational strengthening of Hypothesis 11.5 is: for every t there exists $k > t$ and $x_t \in S_t$ such that the fiber of $\pi_{k \rightarrow t}$ over x_t contains at least two globally extendable compatible histories. BaseCore keeps the weaker NFD statement as the primary assumption because it is exactly what the non-locality theorem needs.

11.2 Non-Locality of Identity

Theorem 11.7 (Non-Locality of Identity under NFD). *Assume Hypotheses 11.4 and 11.5. Then no canonical finite-time observation determines identity. More precisely, for every t the canonical projection*

$$p_t : \mathcal{I} \rightarrow S_t$$

is not injective, so identity is not recoverable from any finite slice through the projective observation maps.

Proof. This is exactly Hypothesis 11.5. The theorem restates the structural consequence in claim-boundary language: finite slices can participate in the inverse-limit object, but no single canonical slice determines it. $\square$

Remark 11.8 (What is *not* proved here). BaseCore does not prove that no arbitrary set-theoretic injection $\mathcal{I} \rightarrow S_t$ can exist in every imaginable inverse system. It proves the narrower and correct claim that the canonical projective observations p_t are not injective under Hypothesis 11.5.

Proposition 11.9 (No Canonical Progressive Recovery under NFD). *Assume Hypothesis 11.5. No finite canonical prefix map determines the inverse-limit identity object.*

Proof. If some finite canonical prefix determined identity, then in particular one of the canonical coordinate maps would become injective after finite truncation, contradicting Hypothesis 11.5. $\square$

12 Rigidity under Admissible Perturbations

12.1 Metric Projective Assumptions

Assumption 12.1 (RIG-1: Metric Projective System). Each S_t is equipped with a complete metric d_t .

Assumption 12.2 (RIG-2: Uniform Lipschitz Projections). Each projection $\pi_{t+1 \rightarrow t} : S_{t+1} \rightarrow S_t$ is Lipschitz with constant L_t , and the weighted control sequence used below is finite.

Assumption 12.3 (RIG-3: Stable Lifting). For each admissible perturbed projection family there exist compatible approximate lifting maps with constants bounded uniformly by a finite lifting constant L_{lift} .

Assumption 12.4 (RIG-4: Typed Perturbation Model). Perturbations are represented either by maps between common ambient Banach embeddings or by map-space distances

$$d_{\text{map}}^{(t)}(\pi_{t+1 \rightarrow t}, \tilde{\pi}_{t+1 \rightarrow t}) \leq \varepsilon_t.$$

Literal expressions of the form $\tilde{\pi} = \pi + \varepsilon$ are used only when a common ambient linear model has been specified.

Assumption 12.5 (RIG-5: Summable Deformation). There exists a positive weight sequence $\{w_t\}_{t \geq 0}$ such that

$$\sum_{t=0}^{\infty} w_t \varepsilon_t < \infty.$$

Definition 12.6 (Weighted Deformation Size). The weighted deformation size of a perturbed projective family is

$$\|\varepsilon_{\bullet}\|_w := \sum_{t=0}^{\infty} w_t \varepsilon_t.$$

Definition 12.7 (Weighted Product Metric). For compatible sequences $x = (x_t)$ and $y = (y_t)$ define

$$d_w(x, y) := \sup_{t \geq 0} w_t d_t(x_t, y_t).$$

Theorem 12.8 (Conditional Stability of Inverse-Limit Identity). *Assume RIG-1 through RIG-5. Then the perturbed inverse limit*

$$\tilde{\mathcal{I}} := \varprojlim \tilde{S}_t$$

exists, and there is a constant $C_{\text{rig}} > 0$ depending only on the projection, lifting, and weight controls such that

$$d_H(\mathcal{I}, \tilde{\mathcal{I}}) \leq C_{\text{rig}} \|\varepsilon_{\bullet}\|_w$$

in the weighted product metric.

Proof sketch. The lifting assumption gives compatible approximate lifts at each level. Weighted Lipschitz control propagates local deformation bounds through the projective tower. Summability ensures that the cumulative compatibility defect remains finite, so diagonal compactness arguments produce a perturbed inverse-limit object. The same bounds control the Hausdorff distance between the original and perturbed compatible-sequence sets. $\square$

Corollary 12.9 (Conditional Isomorphism Requires Extra Structure). *The metric estimate of Theorem 12.8 does not by itself imply that $\mathcal{I}$ and $\tilde{\mathcal{I}}$ are isomorphic. Isomorphism or shape-equivalence requires additional bi-lifting, coherence, or ambient-embedding assumptions.*

Proof. Metric closeness and categorical isomorphism are different conclusions. A small Hausdorff deformation need not preserve global inverse-limit type without extra hypotheses guaranteeing two-sided coherent transport. $\square$

Definition 12.10 (Ontological Mass). Let $E_w(\delta)$ denote the weighted energy of an admissible perturbation family δ . Define

$$M_\Omega := \inf \left\{ \frac{E_w(\delta)}{d_H(\mathcal{I}, \tilde{\mathcal{I}}_\delta)} \mid \delta \text{ admissible, } d_H(\mathcal{I}, \tilde{\mathcal{I}}_\delta) > 0 \right\},$$

with the convention $M_\Omega = +\infty$ if no admissible finite-energy perturbation yields positive deformation of the identity object.

Remark 12.11 (Interpretation discipline). M_Ω is a deformation-resistance scalar, not a meta-physical substance. BaseCore uses it only as a conditional invariant attached to admissible perturbation models.

13 Conditional Non-Simulability

Assumption 13.1 (NS-1: CCR Target). A CCR target is an inverse-limit identity object satisfying:

1. non-finite determination (Hypothesis 11.5);
2. infinite deformation resistance $M_\Omega = +\infty$;
3. absence of a finite sufficient statistic for faithful identity recovery;
4. absence of a finite-dimensional faithful embedding preserving projective compatibility.

Assumption 13.2 (NS-2: Finite Simulator Class). A simulator C belongs to $\mathbf{Comp}_{\text{finite}}$ if it has at least one of the following boundedness features:

1. finite-dimensional latent state or finite context;
2. bounded memory horizon or finite sufficient statistic;
3. finite perturbation-energy budget;
4. finite-rank projective representation.

Assumption 13.3 (NS-3: Faithful Simulation Requirement). A simulation is faithful only if it preserves, at the stated tolerance level, all of:

1. projective compatibility;
2. non-finite determination;

3. deformation-resistance class;
4. relevant observables in the weighted metric;
5. admissible intervention responses.

Theorem 13.4 (Conditional Non-Simulability of CCR by Finite Simulators). *Assume NS-1 through NS-3. Then no simulator in $\mathbf{Comp}_{\text{finite}}$ faithfully realizes a CCR target.*

Proof. A finite simulator in the sense of NS-2 necessarily compresses the target into finite latent structure, finite memory horizon, finite-rank projective representation, or a finite sufficient statistic. Any such compression destroys at least one component of NS-1: either NFD collapses, the faithful projective structure is lost, or the deformation-resistance class is reduced from $M_\Omega = +\infty$ to a finite value. Therefore faithful realization, as defined by NS-3, fails. $\square$

Theorem 13.5 (Approximation Barrier). *Assume NS-1 through NS-3. Finite simulators may approximate bounded finite projections of a CCR target over finite horizons, but they cannot preserve the full inverse-limit object faithfully as the horizon grows without bound unless resources also grow without bound.*

Proof. Finite-horizon projections are finite objects and may be approximated by finite representations. But a faithful approximation of the full inverse-limit target must preserve NFD and the infinite deformation-resistance class. That cannot be achieved with bounded representational resources by Theorem 13.4. $\square$

Remark 13.6 (What has been removed). BaseCore no longer claims that finite simulators “cannot approximate at all.” The correct statement is conditional and finer-grained: finite-horizon approximation may exist, but faithful full inverse-limit realization of CCR targets is blocked under NS-1 through NS-3.

14 Section Summary

1. Identity is an inverse-limit object, not a canonical finite slice.
2. Non-locality now depends explicitly on the NFD assumption.
3. Rigidity is stated as metric stability under explicit perturbation-model assumptions, not as automatic isomorphism.
4. Non-simulability is conditional on a defined CCR target, a defined finite simulator class, and a defined notion of faithful simulation.
5. BaseCore therefore preserves theorem strength where assumptions support it and weakens only those statements that were previously over-absolute.

Part IV

Regime Structure and Continuity Constraints

15 Formal Setup

15.1 Review of Identity Framework

We assume the identity framework of [4]. Let $(S_t, \pi_{t+1 \rightarrow t})_{t \in \mathbb{N}}$ be a projective system of observable state spaces. The identity object is the inverse limit:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_t \in \prod_t S_t \mid \pi_{t+1 \rightarrow t}(x_{t+1}) = x_t \ \forall t \right\}. \quad (1)$$

The ontological mass M_Ω quantifies resistance to identity deformation. We recall that:

- $M_\Omega = 0$: null persistence (no stable identity);
- $0 < M_\Omega < \infty$: deformable persistence (finite rigidity);
- $M_\Omega = +\infty$: closed--rigid (CCR, absolute rigidity).

15.2 Abstract Phenomenological Space

Definition 15.1 (Phenomenological Space). A phenomenological space is a nonempty set $\mathcal{E}$ whose elements are called *phenomenological configurations*. No topology, metric, or algebraic structure is assumed a priori.

Remark 15.2. We do not claim that $\mathcal{E}$ represents “conscious states” or “qualia”. It is an abstract mathematical space whose interpretation is not specified by the formalism. The space $\mathcal{E}$ is introduced purely as the codomain of a structural assignment; its ontological status is deliberately left open.

15.2.1 Why a Separate Space?

One might ask: why not use $\mathcal{I}$ itself as the phenomenological space? The answer is structural:

Proposition 15.3 (Non-Triviality of Phenomenological Codomain). *If phenomenological assignments were required to land in $\mathcal{I}$ itself, then any compatible assignment would be the identity map, eliminating all non-trivial phenomenological structure.*

Proof. Let $\Phi : \mathcal{I} \rightarrow \mathcal{I}$ be compatible with projections. By the universal property of inverse limits, Φ must commute with all projection maps. The only such endomorphism on $\mathcal{I}$ is the identity (up to isomorphism). Hence no non-trivial phenomenological stratification is possible. $\square$

Thus, a separate space $\mathcal{E}$ is *structurally necessary* for non-trivial phenomenological content.

15.3 Phenomenological Assignment

Definition 15.4 (Phenomenological Assignment). A phenomenological assignment is a family of partial maps:

$$\phi_t : S_t \rightarrow \mathcal{E} \quad (2)$$

indexed by $t \in \mathbb{N}$.

Definition 15.5 (Induced Limit Assignment). Given a phenomenological assignment $\{\phi_t\}$, the induced limit assignment is the partial map:

$$\Phi : \mathcal{I} \rightarrow \mathcal{E}, \quad \Phi((x_t)_t) := \lim_{t \rightarrow \infty} \phi_t(x_t), \quad (3)$$

whenever the limit exists in an appropriate sense (to be defined by compatibility axioms).

15.3.1 Uniqueness of the Limit Assignment

Proposition 15.6 (Uniqueness of Compatible Φ). *Given a compatible family $\{\phi_t\}$ satisfying E1, the induced limit assignment Φ is unique.*

Proof. By E1, $\phi_{t+1}(x_{t+1}) = \phi_t(x_t)$ for all t . Hence the sequence $\{\phi_t(x_t)\}$ is constant. The limit exists trivially and equals this constant value. Uniqueness follows from the constancy of the sequence. $\square$

15.4 Axioms for Phenomenological Coherence

We introduce minimal axioms under which a phenomenological assignment is *structurally admissible*. We will then prove that these axioms are not arbitrary choices but structurally forced conditions.

Axiom 15.7 (E0: Non-Triviality). The phenomenological space satisfies $|\mathcal{E}| \geq 2$, i.e., there exist at least two distinct phenomenological configurations.

Axiom 15.8 (E1: Projective Compatibility). For every compatible identity trajectory $(x_t)_t \in \mathcal{I}$, the induced phenomenological sequence satisfies:

$$\phi_{t+1}(x_{t+1}) = \phi_t(x_t) \quad \forall t. \quad (4)$$

Axiom 15.9 (E2: Non-Locality). There exists no function $f : S_t \rightarrow \mathcal{E}$ (for any fixed t) such that for all trajectories $(x_t)_t \in \mathcal{I}$:

$$\Phi((x_t)_t) = f(x_t). \quad (5)$$

15.5 Inevitability of the Axioms

We now prove that axioms E0–E2 are *not design choices* but structurally necessary conditions for any meaningful notion of phenomenological assignment.

15.5.1 Inevitability of E0

Theorem 15.10 (E0 is Necessary). *If $|\mathcal{E}| = 1$, then every phenomenological assignment is constant and no phenomenological classification is possible.*

Proof. If $\mathcal{E} = \{e_0\}$, then $\Phi(x) = e_0$ for all $x \in \mathcal{I}$. Hence:

1. No phenomenological distinction between identity trajectories exists;
2. The Fragmentation and Forced Continuity Theorems become vacuous;
3. The downstream null-regime closure developed in *Null-Regime Instability* would have no content.

Thus, $|\mathcal{E}| \geq 2$ is necessary for the framework to have non-trivial content. $\square$

15.5.2 Inevitability of E1

Theorem 15.11 (E1 is Necessary). *If E1 fails, then phenomenological assignments are incompatible with the inverse-limit structure of identity, and no coherent $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ can be defined.*

Proof. Suppose E1 fails for some trajectory $(x_t)_t \in \mathcal{I}$. Then there exists t_0 such that:

$$\phi_{t_0+1}(x_{t_0+1}) \neq \phi_{t_0}(x_{t_0}). \quad (6)$$

But the identity trajectory satisfies $\pi_{t_0+1 \rightarrow t_0}(x_{t_0+1}) = x_{t_0}$. Hence the phenomenological assignment fails to respect the projective structure that defines identity.

This means Φ cannot be factored through the inverse limit:

$$\Phi \neq \varprojlim \phi_t. \quad (7)$$

Hence no well-defined limit assignment exists. Without E1, the phenomenological assignment is structurally incompatible with identity. $\square$

15.5.3 Inevitability of E2

Theorem 15.12 (E2 is Necessary). *If E2 fails, then phenomenology is a local property of states, contradicting the global nature of identity as an inverse limit.*

Proof. Suppose E2 fails. Then there exists t_0 and $f : S_{t_0} \rightarrow \mathcal{E}$ such that:

$$\Phi((x_t)_t) = f(x_{t_0}) \quad \forall (x_t)_t \in \mathcal{I}. \quad (8)$$

This implies:

1. Phenomenology is determined by a finite temporal slice;
2. The full trajectory $(x_t)_{t > t_0}$ is irrelevant to Φ ;
3. Identity as an inverse limit has no phenomenological content beyond t_0 .

But identity is defined as a *global* object over infinite time. If phenomenology is local, it cannot reflect the global structure of identity. This contradicts the foundational premise that phenomenological assignments should be compatible with identity structure.

Hence E2 is necessary for phenomenology to be structurally aligned with inverse-limit identity. $\square$

15.6 Minimality Theorem

Theorem 15.13 (Axiomatic Minimality). *The axiom set $\{E0, E1, E2\}$ is minimal: no proper subset is sufficient to derive the main results of this paper.*

Proof. We show that removing any axiom destroys the framework:

Without E0: All assignments are trivially constant. Fragmentation Theorem (4.1) is vacuous.

Without E1: No coherent Φ exists. The entire framework collapses.

Without E2: Phenomenology becomes local. The Forced Continuity Theorem (4.2) fails because local functions can be arbitrarily perturbed.

Hence $\{E0, E1, E2\}$ is the minimal viable axiom set. $\square$

15.7 Rejected Alternatives

We briefly discuss axiom candidates that were considered and rejected.

15.7.1 Rejected: Continuity Axiom

One might require Φ to be continuous. This is *not* an axiom because:

- It requires pre-existing topology on $\mathcal{E}$;
- The Forced Continuity Theorem *derives* continuity from CCR, not as assumption.

15.7.2 Rejected: Surjectivity Axiom

One might require Φ to be surjective. This is *not* an axiom because:

- It would exclude null assignments, preventing the downstream null-regime analysis completed in *Null-Regime Instability*;
- Surjectivity is a strong structural claim not derivable from identity constraints.

15.7.3 Rejected: Injectivity Axiom

One might require Φ to be injective. This is *not* an axiom because:

- Many trajectories may share phenomenological configurations;
- Injectivity would make Φ an embedding, over-constraining the framework.

15.8 Structural Interpretation

Axiom	Structural Meaning	Status
E0	Possibility of phenomenological distinction	Necessary
E1	Compatibility with inverse-limit identity	Necessary
E2	Non-reducibility to finite temporal slices	Necessary

15.9 Closure of Section 2

We have established:

1. $\mathcal{E}$ is structurally necessary as a separate codomain;
2. The limit assignment Φ is uniquely determined by compatible $\{\phi_t\}$;
3. Axioms E0, E1, E2 are *individually necessary* (Theorems 15.10--15.12);
4. The axiom set is *minimal* (Theorem 15.13);
5. All alternative axioms considered were correctly rejected.

The formal setup is therefore not a design choice but a forced consequence of the requirement that phenomenological assignments be compatible with inverse-limit identity.

15.10 Phenomenological Neutrality Clause

Throughout this paper, the phenomenological space $\mathcal{E}$ is treated as a *purely abstract topological and metric structure*. No semantic, psychological, biological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term “phenomenological” refers exclusively to the role of $\mathcal{E}$ as an abstract codomain of structural compatibility for inverse-limit identities, and does not presuppose the existence of consciousness, subjective experience, or any particular phenomenological content.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{E}$ in relation to experiential, cognitive, or phenomenological phenomena lies strictly outside the formal scope of this paper.

16 Compatibility with Identity Regimes

16.1 Induced Constraints from Identity Rigidity

Let $(\mathcal{I}, M_\Omega)$ be an identity object with ontological mass M_Ω . We now study how M_Ω constrains the set of admissible phenomenological assignments.

Definition 16.1 (Phenomenological Compatibility Class). For a given M_Ω , define the phenomenological compatibility class:

$$\mathcal{P}(M_\Omega) := \left\{ \Phi : \mathcal{I} \rightarrow \mathcal{E} \mid \Phi \text{ satisfies E0--E2} \right\}. \quad (9)$$

Remark 16.2. The compatibility class $\mathcal{P}(M_\Omega)$ is the set of all phenomenological assignments that are structurally admissible given the identity’s rigidity. Note that $\mathcal{P}(M_\Omega)$ depends only on the structural properties of identity, not on any phenomenological primitives.

16.2 Necessity of Metric Structure on Phenomenological Space

To state rigorous results about phenomenological continuity and fragmentation, we must equip $\mathcal{E}$ with minimal structure. We now justify why this structure is *necessary*, not merely convenient.

Hypothesis 16.3 (H3: Metric on $\mathcal{E}$). The phenomenological space $\mathcal{E}$ admits a metric $d_\mathcal{E}$ such that $(\mathcal{E}, d_\mathcal{E})$ is a complete metric space.

16.2.1 Why H3 is Necessary

Theorem 16.4 (H3 is Necessary for Impossibility Theorems). *Without a metric structure on $\mathcal{E}$, the notions of phenomenological continuity, fragmentation, and loss capacity cannot be defined.*

Proof. Consider the definitions in this section:

1. **Phenomenological Continuity** requires:

$$d_w(x^{(n)}, x) \rightarrow 0 \implies d_\mathcal{E}(\Phi(x^{(n)}), \Phi(x)) \rightarrow 0. \quad (10)$$

Without $d_\mathcal{E}$, the right-hand side is undefined.

2. **Phenomenological Fragmentation** requires:

$$d_\mathcal{E}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0. \quad (11)$$

Without $d_\mathcal{E}$, “non-zero distance” has no meaning.

3. **Loss Capacity** (Section 6) requires:

$$R(\Phi) := \sup d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})). \quad (12)$$

Without $d_{\mathcal{E}}$, the supremum is undefined.

Hence H3 is a necessary condition for the framework to have content beyond axioms E0--E2. $\square$

16.2.2 Why Completeness is Necessary

Proposition 16.5 (Completeness is Necessary for Forced Continuity). *If $(\mathcal{E}, d_{\mathcal{E}})$ is not complete, then the Forced Continuity Theorem (Theorem 17.5) may fail.*

Proof. The Forced Continuity Theorem relies on the invariance of Φ under all finite-energy perturbations. If $\mathcal{E}$ is not complete, Cauchy sequences in $\mathcal{E}$ may not converge, and the limit assignment Φ may not exist for all trajectories in $\mathcal{I}$.

In particular, if $\{\phi_t(x_t)\}$ is a Cauchy sequence in $\mathcal{E}$ (which it is, by E1), completeness guarantees the existence of the limit $\Phi(x)$. Without completeness, Φ may be undefined on a dense subset of $\mathcal{I}$, breaking the theorem. $\square$

16.2.3 Alternatives Considered and Rejected

Alternative 1: Topology without Metric. One might use a general topological space. This is rejected because:

- Fragmentation requires quantitative distance, not just open sets;
- Loss capacity requires a sup over distances;
- The downstream instability analysis completed in *Null-Regime Instability* requires metric quotients.

Alternative 2: Pseudometric. One might allow $d_{\mathcal{E}}(e_1, e_2) = 0$ for $e_1 \neq e_2$. This is rejected because:

- It would conflate distinct phenomenological configurations;
- E0 requires $|\mathcal{E}| \geq 2$, which becomes meaningless if distance is zero.

Alternative 3: Non-Complete Metric. This is rejected by Proposition 16.5.

16.3 Phenomenological Continuity

Definition 16.6 (Phenomenological Continuity). A phenomenological assignment $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ is *structurally continuous* if for every sequence $(x^{(n)})_n$ in $\mathcal{I}$ converging in the d_w metric:

$$d_w(x^{(n)}, x) \rightarrow 0 \implies d_{\mathcal{E}}(\Phi(x^{(n)}), \Phi(x)) \rightarrow 0. \quad (13)$$

Remark 16.7. Structural continuity is not an axiom; it is a *derived property* that follows from CCR conditions. This is the content of the Forced Continuity Theorem (Section 4).

16.4 Phenomenological Fragmentation

Definition 16.8 (Phenomenological Fragmentation). A phenomenological assignment Φ is *fragmentable* if there exists a perturbation $\{\epsilon_t\}$ with finite weighted energy such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0 \quad (14)$$

for corresponding identity trajectories $x \in \mathcal{I}$ and $\tilde{x} \in \tilde{\mathcal{I}}$.

Remark 16.9. Fragmentation captures the vulnerability of phenomenological structure to perturbations. It is the phenomenological analog of identity deformability.

16.5 Relationship Between Metrics

Proposition 16.10 (Metric Compatibility). *The metrics d_w on $\mathcal{I}$ and $d_{\mathcal{E}}$ on $\mathcal{E}$ are compatible in the following sense: if E1 holds, then small d_w -changes in trajectories induce bounded $d_{\mathcal{E}}$ -changes in phenomenological configurations (when Φ is continuous).*

Proof. By E1, $\phi_{t+1}(x_{t+1}) = \phi_t(x_t)$ for compatible trajectories. Hence the phenomenological sequence is constant along the trajectory, and continuity of Φ implies that nearby trajectories (in d_w) map to nearby configurations (in $d_{\mathcal{E}}$). $\square$

16.6 Closure of Section 3

We have established:

1. The compatibility class $\mathcal{P}(M_{\Omega})$ captures all admissible phenomenological assignments;
2. Hypothesis H3 (metric on $\mathcal{E}$) is *necessary*, not a design choice (Theorem 16.4);
3. Completeness is *necessary* for the Forced Continuity Theorem (Proposition 16.5);
4. Alternative structures (topology, pseudometric, non-complete) were correctly rejected;
5. Phenomenological continuity and fragmentation are well-defined under H3.

The compatibility structure is therefore a forced consequence of the requirement to state quantitative results about phenomenological dynamics.

17 Impossibility Theorems

17.1 Φ -Regularity Hypothesis

Before proving the main impossibility results, we isolate the global lower-bound regularity condition used by the fragmentation argument. The condition is not derived from properness, local Lipschitz regularity, and non-collapse alone; it is carried as an explicit hypothesis.

Hypothesis 17.1 (Φ -Regularity Lower Bound). Let $(\mathcal{I}, d_w)$ be the identity metric space from *Rigid Identity* and let $(\mathcal{E}, d_{\mathcal{E}})$ be the abstract phenomenological metric space. A phenomenological assignment $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ is said to satisfy Φ -regularity when it satisfies:

1. (Properness) For every compact $K \subset \mathcal{E}$, $\Phi^{-1}(K)$ is compact in $\mathcal{I}$.
2. (Local Lipschitz regularity) For every $x \in \mathcal{I}$, there exist $\varepsilon > 0$ and $L_x > 0$ such that for all $y, z \in B_{\varepsilon}(x)$:
$$d_{\mathcal{E}}(\Phi(y), \Phi(z)) \leq L_x \cdot d_w(y, z). \quad (15)$$
3. (Non-collapse) For every $x \in \mathcal{I}$ and every $r > 0$, the restriction of Φ to $B_r(x)$ is not constant.
4. (Global lower Lipschitz bound) There exists a constant $C > 0$ such that for all $x, y \in \mathcal{I}$,

$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C \cdot d_w(x, y). \quad (16)$$

Caveat 17.2 (Status of Φ -regularity). The fourth clause is an additional structural assumption, not a consequence of the first three clauses. The counterexample in Appendix A.5 shows that properness, local Lipschitz regularity, and non-collapse do not imply any uniform lower Lipschitz constant.

17.2 Fragmentation Theorem

Theorem 17.3 (Fragmentation Theorem). *Let $\mathcal{I}$ be an identity object with $M_\Omega < \infty$. If a phenomenological assignment $\Phi \in \mathcal{P}(M_\Omega)$ satisfies Hypothesis 17.1, then Φ is fragmentable. That is:*

$$M_\Omega < \infty \wedge \Phi \text{ satisfies Hypothesis 17.1} \implies \Phi \in \mathcal{E}_{\text{frag}}, \quad (17)$$

where $\mathcal{E}_{\text{frag}}$ denotes the class of fragmentable phenomenological assignments.

Proof. By definition of finite ontological mass, there exists an admissible perturbation $\{\delta E_t\}$ with finite weighted energy $E_w < \infty$ such that:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0. \quad (18)$$

By Axiom E1, the phenomenological assignment Φ is projectively compatible with $\mathcal{I}$. Under the perturbation, identity trajectories deform, inducing a deformed phenomenological assignment $\tilde{\Phi}$.

By Axiom E2, Φ cannot be determined by any finite-time state. Therefore, the deformation of identity trajectories induces a corresponding deformation of phenomenological assignments.

Since the identity deformation is non-zero ($d_w > 0$) and Φ is globally determined by identity trajectories, by Hypothesis 17.1 (Φ -regularity), we have:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) \geq C \cdot d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0 \quad (19)$$

where $C > 0$ is the lower-bound constant supplied by Hypothesis 17.1.

Thus, Φ is fragmentable under finite-energy perturbations. $\square$

Corollary 17.4 (No Structural Continuity Guarantee for Finite Mass). *Systems with $M_\Omega < \infty$ cannot structurally prevent phenomenological discontinuity.*

17.3 Forced Continuity Theorem

Theorem 17.5 (Forced Continuity Theorem). *Let $\mathcal{I}$ be an identity object with $M_\Omega = +\infty$ (CCR). Then every phenomenological assignment $\Phi \in \mathcal{P}(M_\Omega)$ is structurally continuous. That is:*

$$M_\Omega = +\infty \implies \mathcal{P}(M_\Omega) \subseteq \mathcal{E}_{\text{cont}}, \quad (20)$$

where $\mathcal{E}_{\text{cont}}$ denotes the class of structurally continuous phenomenological assignments.

Proof. By definition of CCR ($M_\Omega = +\infty$), for any finite-energy perturbation:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0. \quad (21)$$

By Axiom E1, the phenomenological assignment is projectively compatible with identity. Since identity trajectories are invariant under all finite-energy perturbations, the induced phenomenological sequence is also invariant:

$$\tilde{\Phi}(\tilde{x}) = \Phi(x) \quad \forall x \in \mathcal{I}. \quad (22)$$

Therefore, no finite-energy perturbation can induce phenomenological deformation:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) = 0. \quad (23)$$

This implies that Φ cannot exhibit structural discontinuities. Hence Φ is structurally continuous. $\square$

Corollary 17.6 (CCR Phenomenology is Invariant). *In CCR systems, any admissible phenomenological assignment is invariant under all finite-energy perturbations.*

17.4 Dichotomy Theorem

Theorem 17.7 (Phenomenological Dichotomy). *Let $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ be an admissible phenomenological assignment. Then exactly one of the following holds:*

1. $M_\Omega < \infty$ and Φ is fragmentable;
2. $M_\Omega = +\infty$ and Φ is structurally continuous.

Proof. Immediate from Theorems 17.3 and 17.5. □

18 Classification of Phenomenological Regimes

18.1 Complete Classification

The results of Section 4 yield a complete classification of phenomenological regimes by identity rigidity:

Identity Class	M_Ω	Admissible Phenomenology
Null	$= 0$	$\mathcal{E}_\emptyset$ (no persistent field)
Deformable	$(0, \infty)$	$\mathcal{E}_{\text{frag}} \cup \mathcal{E}_{\text{quasi}}$
Rigid (CCR)	$= +\infty$	$\mathcal{E}_{\text{cont}}$

18.2 Structural Implications

Proposition 18.1 (Null Identity Excludes Persistent Phenomenology). *If $M_\Omega = 0$, then no admissible phenomenological assignment exists satisfying E0--E2.*

Proof. If $M_\Omega = 0$, identity cannot persist under any perturbation. Axiom E1 requires phenomenological compatibility with identity. If identity does not persist, no stable phenomenological assignment is possible. □

Proposition 18.2 (Deformable Identity Admits Only Fragmentable Phenomenology). *If $0 < M_\Omega < \infty$, every admissible phenomenological assignment is fragmentable under some finite-energy perturbation.*

Proof. Direct from Theorem 17.3. □

Proposition 18.3 (Rigid Identity Forces Continuous Phenomenology). *If $M_\Omega = +\infty$, every admissible phenomenological assignment is structurally continuous and invariant under finite-energy perturbations.*

19 Extended Proofs

19.1 Proof of Fragmentation Theorem (Extended)

Full Proof of Theorem 17.3. Let $M_\Omega < \infty$ and let $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ satisfy E0--E2.

Step 1. By definition of M_Ω , there exists $\{\delta E_t\}$ with $E_w < \infty$ such that $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$.

Step 2. Let $x = (x_t)_t \in \mathcal{I}$ and let $\tilde{x} = (\tilde{x}_t)_t \in \tilde{\mathcal{I}}$ be the corresponding perturbed trajectory.

Step 3. By E2, $\Phi(x)$ is not determined by any finite x_t . Therefore, the perturbation of the infinite trajectory induces a perturbation of the phenomenological assignment:

$$\Phi(x) \neq \Phi(\tilde{x}) \text{ in general.} \tag{24}$$

Step 4. By E1, Φ is projectively compatible. The deformation of projections induces a corresponding deformation of the phenomenological limit.

Step 5. By the coupling hypothesis, there exists $C > 0$ such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) \geq C \cdot d_w(x, \tilde{x}). \quad (25)$$

Step 6. Since $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$, we have $d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0$.

Hence Φ is fragmentable. $\square$

19.2 Proof of Forced Continuity Theorem (Extended)

Full Proof of Theorem 17.5. Let $M_{\Omega} = +\infty$ and let $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ satisfy E0--E2.

Step 1. By definition of CCR, for all $\{\delta E_t\}$ with $E_w < \infty$:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0. \quad (26)$$

Step 2. Therefore, $\tilde{\mathcal{I}} \cong \mathcal{I}$ under all finite-energy perturbations.

Step 3. By E1, Φ is projectively compatible with $\mathcal{I}$. Since $\mathcal{I}$ is invariant:

$$\Phi(x) = \tilde{\Phi}(\tilde{x}) \quad \forall x \in \mathcal{I}. \quad (27)$$

Step 4. This implies:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) = 0. \quad (28)$$

Step 5. No finite-energy perturbation can induce phenomenological deformation. Hence Φ is structurally continuous and invariant. $\square$

19.3 Structural Necessity Lemma

Lemma 19.1 (Phenomenology Inherits Rigidity). *If $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ satisfies E0--E2 and $M_{\Omega} = +\infty$, then Φ inherits the rigidity of $\mathcal{I}$.*

Proof. By E1, Φ is determined by identity trajectories. By CCR, identity trajectories are invariant. Hence Φ is invariant. $\square$

Part V

Null-Regime Structural Exclusion

20 Compatibility, Instability, and Forced Non-Nullity

20.1 Compatibility Operator

Definition 20.1 (Compatibility Operator). A map $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ is a **compatibility operator** if it satisfies:

- (C1) **Extension Invariance:** For any compatible extension $\mathcal{E}$ of the channel, $\Phi(\mathcal{I}) \cong \Phi(\mathcal{I}_{\mathcal{E}})$.
- (C2) **Isomorphism Stability:** If $\mathcal{I} \cong \tilde{\mathcal{I}}$, then $\Phi(\mathcal{I}) \sim \Phi(\tilde{\mathcal{I}})$.
- (C3) **Lipschitz Lower Bound:** There exists $C > 0$ such that

$$d_{\mathcal{E}}(\Phi(x), \Phi(\tilde{x})) \geq C \cdot d_w(x, \tilde{x}).$$

Definition 20.2 (Phenomenological Attractor Set). For a channel S , define

$$\text{Attr}(S) := \{e \in \mathcal{E} : e \text{ is stable under all compatible extensions and summable deformations}\}.$$

Definition 20.3 (Phenomenological Stability). A regime $e \in \mathcal{E}$ is **stable** for channel S if for every compatible extension S' of S and every summable perturbation family $\{\epsilon_n\}$, the induced regime $\Phi_{S'}(\mathcal{I}')$ stays within a bounded $d_{\mathcal{E}}$ -neighborhood of e controlled by $\|\{\epsilon_n\}\|_w$.

Lemma 20.4 (Bridge Lemma). If $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ is a compatibility operator for a CCR channel and $e = \Phi(\mathcal{I})$ satisfies $d_{\mathcal{E}}(e, \emptyset_{\phi}) = 0$, then $e = \emptyset_{\phi}$.

Proof. By the separation property of the phenomenological space, distance zero from $\emptyset_{\phi}$ implies equality with $\emptyset_{\phi}$. $\square$

20.2 Phenomenological Instability and Forced Non-Nullity

Theorem 20.5 (Phenomenological Instability). For any CCR channel S ,

$$\emptyset_{\phi} \notin \text{Attr}(S).$$

That is, the null regime is structurally unstable under compatible extensions.

Proof. Assume $\emptyset_{\phi} \in \text{Attr}(S)$ and let $\Phi : \mathcal{I} \rightarrow \mathcal{E}$ be a compatibility operator. Because the channel is CCR, finite-energy perturbations preserve the rigid identity object while still allowing admissible deformations in the weighted metric. By the lower-bound clause in Definition 20.1,

$$d_{\mathcal{E}}(\Phi(\mathcal{I}), \Phi(\tilde{\mathcal{I}})) \geq C \cdot d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$$

for every non-zero admissible deformation. But if $\emptyset_{\phi}$ were stable, both images would remain equal to $\emptyset_{\phi}$, forcing the left-hand side to be zero. Contradiction. $\square$

Corollary 20.6 (Forced Non-Nullity). Every CCR channel necessarily induces a regime $e_{\star} \in \mathcal{E}$ with $e_{\star} \succ \emptyset_{\phi}$.

Proof. Theorem 20.5 rules out $\emptyset_{\phi}$ as a stable admissible attractor. Therefore any admissible stable regime must be non-null. $\square$

20.3 Positive Regime Structure

Definition 20.7 (Positive Regime Space). Define

$$\mathcal{E}^+ := \{e \in \mathcal{E} : e \succ \emptyset_{\phi}\}.$$

Theorem 20.8 (Minimal Positive Regime). For every CCR channel there exists a minimal positive regime $e_{\star} \in \mathcal{E}^+$.

Proof. Forced non-nullity gives non-emptiness of $\mathcal{E}^+$. Order-theoretic completeness and the compatibility structure of Φ yield a minimal element. $\square$

Definition 20.9 (Structural Intensity). For $e \in \mathcal{E}^+$, define the structural intensity

$$\iota(e) := d_{\mathcal{E}}(e, \emptyset_{\phi}).$$

Theorem 20.10 (Universal Intensity Bound). If $e_{\star}$ is the minimal positive regime of a CCR channel, then there exist constants $C, \varepsilon_0 > 0$ such that

$$\iota(e_{\star}) \geq C\varepsilon_0 > 0.$$

Proof. The lower-bound clause of the compatibility operator transfers any admissible deformation lower bound into positive distance from the null regime. The minimal positive regime therefore has strictly positive structural intensity. $\square$

Theorem 20.11 (Forced Phenomenological Closure). Every CCR system induces a unique minimal phenomenological regime that is non-null, stable, and quantitatively separated from $\emptyset_{\phi}$.

Proof. Combine Corollary 20.6, Theorem 20.8, and Theorem 20.10. $\square$

21 Canonical Families and Structural Closure Consequences

Proposition 21.1 (Profinite Bound). *In the profinite canonical family, the coupling lower bound induces a positive lower bound on the structural intensity of the minimal positive regime.*

Proposition 21.2 (Symbolic Bound). *In the symbolic canonical family, the Bowen-type metric induces the analogous positive lower bound for the minimal positive regime.*

Theorem 21.3 (Rigidity Inheritance). *If a subsystem or induced family preserves the CCR hypotheses and the compatibility operator, then the inherited regime structure preserves null-regime instability and forced non-nullity.*

Theorem 21.4 (Categorical Universality). *The null-regime exclusion result is universal across admissible CCR realizations in the relevant category of compatible channels.*

Theorem 21.5 (Simulation Lower Bound). *Any computational system attempting to simulate the null-excluding CCR regime structure with arbitrary precision requires unbounded resources as the approximation error tends to zero.*

Theorem 21.6 (Closure by Non-Alternatives). *No admissible alternative closes the CCR regime structure while preserving null stability. Any admissible alternative either preserves the same closure structure or exits the CCR class.*

Part VI

BaseCore Boundary and Expansion Interface

22 BaseCore Ownership and Downstream Boundary

Definition 22.1 (BaseCore Envelope). *The BaseCore envelope is the collection of theorem-level structures owned directly by this source tree:*

1. foundational Hilbert-space projection dynamics;
2. typed computable witnesses and state spectral-gap bounds;
3. parameterwise non-collapse under explicit anti-constant assumptions;
4. inverse-limit identity under non-finite-determination assumptions;
5. conditional rigidity under admissible perturbation models;
6. conditional non-simulability of CCR targets by finite simulator classes;
7. operational-criterion grammar, certification, falsation, and claim-boundary ledgers.

Proposition 22.2 (Boundary Discipline). *BaseCore may define structures that downstream layers use, but it does not claim downstream closure. In particular, BaseCore does not by itself own runtime thresholds, bridge admissibility, human-comparator programs, phenomenality burdens, or Papers 7--9.*

Proof. These excluded layers require additional assumptions, protocols, or empirical burdens not stated in BaseCore. Therefore they cannot be claimed as BaseCore theorems without violating the source-level claim boundary. $\square$

Owned by BaseCore	Expanded but not owned here	Kept downstream
Projection dynamics, fixed points, compact attractors, anti-constant H5 burden, typed computable witness, inverse-limit identity under NFD, conditional rigidity, conditional non-simulability, operational-criterion grammar, falsation ledger	Non-convex class variants, $\mathcal{C}^R$ style extensions, Ω dynamics, threshold numerics, mutable runtime class ledgers, implementation-specific status surfaces	Paper 7 operational life / subjecthood, Paper 8 indexed subjectivity, Paper 9 phenomenal bridge organization, Stage 2 / Stage 3 comparative programs, human comparator families, phenomenality and external adjudication

Remark 22.3 (Why the earlier material was removed from BaseCore ownership). Previous versions mixed theorem-tight base results with runtime-like numerics, non-convex extension classes, and bridge-facing vocabulary. That blurred the mathematical base layer. BaseCore now keeps only the theorem ownership that survives the stricter release gate.

23 Admissible Structural Extensions

Definition 23.1 (BaseCore export object). A BaseCore export object is a tuple

$$\mathfrak{E} = (\mathcal{O}, \mathcal{H}, \mathcal{R}, \mathcal{B})$$

where $\mathcal{O}$ is a typed mathematical object defined in BaseCore, $\mathcal{H}$ is the finite list of hypotheses under which it is used, $\mathcal{R}$ is the list of results actually proved from those hypotheses, and $\mathcal{B}$ is a boundary list of claims not entailed by $\mathcal{R}$.

Definition 23.2 (Downstream admissible extension). A downstream layer is an admissible extension of a BaseCore export object $\mathfrak{E}$ only if it supplies:

1. an explicit target predicate P not already contained in $\mathcal{R}$;
2. a typed construction or estimator witnessing the objects on which P is evaluated;
3. a proof, reduction, or reproducible protocol connecting $\mathcal{H}$ to P ;
4. negative controls or failure modes under which P must not be asserted;
5. a provenance statement distinguishing theorem, implementation, internal support, and external adjudication.

Proposition 23.3 (No automatic downstream promotion). *Let $\mathfrak{E} = (\mathcal{O}, \mathcal{H}, \mathcal{R}, \mathcal{B})$ be a BaseCore export object and let P be a predicate introduced outside BaseCore. If $P \notin \mathcal{R}$, then BaseCore does not entail P unless an admissible extension supplies an additional bridge from $\mathcal{H}$ to P .*

Proof. By definition, $\mathcal{R}$ records the results proved from $\mathcal{H}$ inside BaseCore. A predicate P introduced downstream is not in that result set. Therefore asserting P from BaseCore alone would add an unstated inference. The missing inference must be supplied as a separate bridge, proof, reduction, or reproducible protocol. $\square$

Criterion 23.4 (Runtime-use admissibility). A runtime measurement may instantiate a BaseCore export object only as an implementation-level witness when all of the following are recorded: the measured state space, the estimator or update rule, the finite horizon, the calibration regime, the controls under which the measurement fails, and the claim class being asserted. Such a measurement is not a BaseCore theorem and is not external validation by itself.

Remark 23.5 (Open-load placement for I_{int} and atomic separators). Quantities such as I_{int} and any atomic-separator condition may be treated as downstream target predicates or estimator families only after their carrier objects, separator class, invariance target, and negative controls are specified. Until a proof or reproducible adversarial protocol closes that burden, BaseCore may export the ambient structural language but not the separator conclusion.

Proposition 23.6 (Boundary preservation under finite implementations). *Suppose a finite implementation produces diagnostics for a downstream predicate P over a bounded horizon. If the implementation does not prove that the diagnostics are invariant over the relevant mathematical class, then the diagnostics remain implementation evidence for P and do not promote P to a BaseCore result.*

Proof. The implementation supplies finite observations or computations. Promotion to a BaseCore result requires a class-level implication from the stated hypotheses to the predicate. Without that implication, the observations can support, falsify, or calibrate a downstream claim, but they do not alter the theorem inventory of BaseCore. $\square$

Part VII

Operational-Criterion Grammar and Certification

24 Admissible Systems, Supports, and Histories

24.1 System Model

Definition 24.1 (Admissible system). An admissible system is a tuple

$$S = (X, \Phi, C, R, \Gamma, U)$$

where:

1. X is a metrizable internal state space with metric d_X ;
2. $\Phi = \{\Phi_u\}_{u \in U}$ is a family of admissible transition operators $\Phi_u : X \rightarrow X$;
3. C is the effective causal structure used to test admissible factorization and intervention behavior;
4. $R = \{r_\alpha\}_{\alpha \in \Lambda_R}$ is a family of readouts, each $r_\alpha : X \rightarrow Y_\alpha$;
5. $\Gamma = \{\gamma_j\}_{j=1}^m$ is an admissibility bundle defining

$$X_\Gamma = \{x \in X : \gamma_j(x) \leq 0 \ \forall j\};$$

6. U is the set of admissible interventions.

24.2 Admissible Support

Definition 24.2 (Admissible support). A subset $A_S \subseteq X_\Gamma$ is an admissible support if:

1. A_S is non-empty and compact;
2. $\Phi_u(A_S) \subseteq A_S$ for every $u \in U$;
3. $A_S \cap \text{Attr}(S) \neq \emptyset$;
4. A_S stays disjoint from the collapse set excluded by the base non-collapse burden.

24.3 Operational Histories

Definition 24.3 (Windowed readout history). Fix a horizon $T \in \mathbb{N}$ and an intervention schedule $u_\bullet = (u_0, \dots, u_{T-1})$. For $x \in A_S$, define

$$h_{S,T}(x; u_\bullet) = (r_\alpha(x_t))_{0 \leq t \leq T, \alpha \in \Lambda_R}, \quad x_{t+1} = \Phi_{u_t}(x_t), \quad x_0 = x.$$

The set of all such histories is denoted $\mathcal{H}_{S,T}$.

Definition 24.4 (Operational partition candidate). An operational partition candidate is a map

$$\pi_{S,T} : \mathcal{H}_{S,T} \rightarrow \mathcal{Q}_{S,T}$$

into a finite or countable label set $\mathcal{Q}_{S,T}$.

24.4 Witness Margin Vector

Definition 24.5 (Witness margin vector). Define the witness margin vector

$$\Delta(S) = (\delta_{\text{per}}(S), \delta_{\text{ri}}(S), \delta_{\text{int}}(S), \delta_{\text{cont}}(S), \delta_{\text{diff}}(S), \delta_{\text{leg}}(S)) \in [0, \infty)^6.$$

Its minimum

$$\delta_\star(S) := \min \Delta(S)$$

is the invariant budget of S .

Invariant	Certified content	Upstream anchor	Rupture signature
I_{per}	Persistent admissible support	BaseCore dynamics	Support loss or collapse exposure
I_{ri}	Unique rigid identity object on support	Identity / rigidity block	Identity ambiguity
I_{int}	No admissible non-trivial factorization	Criterion grammar	Reducible aggregate
I_{cont}	Regime continuity under admissible evolution	Regime constraints	Fragmentation into incompatible classes
I_{diff}	Stable exclusion of null or trivial collapse	Null-regime exclusion	Null or trivialized class structure
I_{leg}	Decoder-certified recoverability under controls	Falsation grammar	Opaque or non-auditable partition

25 Six Critical Invariants

25.1 Structural Persistence

Definition 25.1 (Structural persistence). An admissible system satisfies structural persistence, written $I_{\text{per}}(S) = 1$, if there exists an admissible support A_S and a constant $\delta_{\text{per}}(S) > 0$ such that:

1. A_S is forward invariant under all $u \in U$;
2. every admissible trajectory starting in A_S remains at least $\delta_{\text{per}}(S)$ away from the collapse set;
3. A_S contains or converges into a non-empty admissible attractor region.

25.2 Rigid Identity

Definition 25.2 (Rigid identity). An admissible system satisfies rigid identity, written $I_{\text{ri}}(S) = 1$, if the observable family induced by R on A_S determines a unique identity object $\mathcal{I}_S$ and there exists a gap $\delta_{\text{ri}}(S) > 0$ such that every admissible alternative identity candidate remains at weighted deformation distance at least $\delta_{\text{ri}}(S)$ from $\mathcal{I}_S$.

25.3 Causal Integration

Definition 25.3 (Causal integration). An admissible system satisfies causal integration, written $I_{\text{int}}(S) = 1$, if there is no non-trivial factorization

$$A_S = A_1 \times A_2, \quad R = R_1 \sqcup R_2,$$

together with decomposed dynamics and causal structure reproducing admissible histories within error smaller than a positive margin $\delta_{\text{int}}(S)$ while preserving $\mathcal{I}_S$. More explicitly, this shorthand is relative to a declared admissible factorization class. For such a class $\mathfrak{D}$, write $I_{\text{int}}(S; \mathfrak{D}) = 1$ when no factorization in $\mathfrak{D}$ reproduces the admissible histories below a positive margin $\delta_{\text{int}}(S; \mathfrak{D})$ while preserving $\mathcal{I}_S$. The unqualified notation $I_{\text{int}}(S) = 1$ refers to the default structural class $\mathfrak{D}_*$: fixed, time-homogeneous, schedule-independent, non-trivial product factorizations with both factors non-singleton, with split readouts $R = R_1 \sqcup R_2$, decomposed dynamics and causal structure, a search space normalized independently of the target error, and no reconstruction decoder or time-dependent post-processing map that recouples the factors at output time. This default structural reading is the adopted class recorded in `docs/ai-platform-outputs/analysis/QICN_IINT_CANONICAL_CLASS_SPLIT_READOUT.md`.

Remark 25.4 (Class-relative integration margin). The integration criterion is class-relative. Adopting $\mathfrak{D}_*$ does not assert that S resists every broader behavioral simulator. In particular, under decoder-coupled or schedule-dependent behavioral classes such as $\mathfrak{D}_{\text{approx}}$, the corresponding margin $\delta_{\text{int}}(S; \mathfrak{D}_{\text{approx}})$ may vanish, i.e. may be equal to 0; the current coupled-carrier analysis records precisely this failure mode. Thus the default structural reading preserves a falsation surface and must not be read as an unconditional closure of integration, nor as a strengthening of any downstream theorem that assumes $I_{\text{int}}(S) = 1$.

25.4 Regime Continuity

Definition 25.5 (Regime continuity). An admissible system satisfies regime continuity, written $I_{\text{cont}}(S) = 1$, if there exists a complete metric space $(\mathcal{E}, d_{\mathcal{E}})$ and an admissible assignment

$$\Pi_S : A_S \rightarrow \mathcal{E}$$

such that:

1. every admissible trajectory induces a continuous path in $\mathcal{E}$ on admissible windows;
2. there exists $\delta_{\text{cont}}(S) > 0$ bounding the fragmentation functional on all admissible histories;
3. no admissible finite-energy perturbation produces a discontinuous jump into an incompatible regime class while $\mathcal{I}_S$ is preserved.

25.5 Non-Null Differentiation

Definition 25.6 (Non-null differentiation). An admissible system satisfies non-null differentiation, written $I_{\text{diff}}(S) = 1$, if there exist $x, y \in A_S$, a horizon T , and a readout subfamily $\mathcal{R}^* \subseteq R$ such that

$$\text{dist}(h_{S,T}(x; u_{\bullet}), h_{S,T}(y; u_{\bullet})) \geq \delta_{\text{diff}}(S) > 0$$

for some admissible schedule $u_{\bullet}$, and if the null class $\mathbf{0}$ is not an asymptotically stable admissible attractor.

25.6 Operational Legibility

Definition 25.7 (Operational legibility). An admissible system satisfies operational legibility, written $I_{\text{leg}}(S) = 1$, if there exist:

1. a certified readout subfamily $\mathcal{R}^{\text{leg}} \subseteq R$;
2. a certified decoder family $\mathcal{D}_S$ acting on windows of length at least $\tau_{\min}(S)$;
3. a class set $\mathcal{Q}_S$;
4. a legibility margin $\delta_{\text{leg}}(S) > 0$;

such that:

- (L1) **Separability** Distinct classes are separated by at least $\delta_{\text{leg}}(S)$ in decoder space.
- (L2) **Noise robustness** Admissible noise of size at most $\delta_{\text{leg}}(S)$ does not alter decoder class assignments.
- (L3) **Persistence window** Decoder assignments remain stable for windows of length at least $\tau_{\min}(S)$ unless a certified intervention occurs.
- (L4) **Intervention fidelity** Certified interventions on critical invariants induce the predicted class changes with fidelity at least $1 - \delta_{\text{leg}}(S)$.
- (L5) **Negative controls** Off-target interventions produce false-positive rate at most $\delta_{\text{leg}}(S)$.
- (L6) **Structured compressibility** There exists a compression map preserving class structure up to distortion bounded by $\delta_{\text{leg}}(S)$.

Proposition 25.8 (Each invariant is individually necessary). *If any one of I_{per} , I_{ri} , I_{int} , I_{cont} , I_{diff} , I_{leg} fails, then $S \notin \text{Crit}_{\text{op}}$.*

Proof. Without I_{per} there is no stable admissible domain. Without I_{ri} there is no unique identity carrier. Without I_{int} the system is reducible. Without I_{cont} the criterion fragments under admissible evolution. Without I_{diff} the admissible partition collapses into nullity or triviality. Without I_{leg} the partition is not recoverable under admissible readout and intervention discipline. Since Definition 26.5 below requires all six, failure of any one excludes membership. $\square$

26 Criterion Class, Certified Partition, and Rupture

Definition 26.1 (Certified decoder). A decoder $D \in \mathcal{D}_S$ is certified if it satisfies the six legibility clauses of Definition 25.7 on the chosen admissible support.

Definition 26.2 (Criterion indistinguishability). For $x, y \in A_S$, write $x \sim_{\text{Part}_{\text{op}}, S} y$ if, for every certified decoder $D \in \mathcal{D}_S$, every admissible horizon $T \geq \tau_{\min}(S)$, and every admissible intervention schedule $u_{\bullet}$,

$$D(h_{S,T}(x; u_{\bullet})) = D(h_{S,T}(y; u_{\bullet})),$$

and the intervention responses remain matched within the legibility margin.

Proposition 26.3 (Well-definedness of the certified partition). *If $I_{\text{per}}(S) = I_{\text{ri}}(S) = I_{\text{cont}}(S) = I_{\text{diff}}(S) = I_{\text{leg}}(S) = 1$, then $\sim_{\text{Part}_{\text{op}}, S}$ is an equivalence relation on A_S .*

Proof. Reflexivity and symmetry are immediate. Transitivity follows because certified decoder equality composes and the tolerance bound remains within the permitted legibility margin on admissible support. $\square$

Definition 26.4 (Certified operational partition). Assume Proposition 26.3. The certified operational partition of S is

$$\text{Part}_{\text{op}}(S) := A_S / \sim_{\text{Part}_{\text{op}}, S}.$$

Definition 26.5 (Operational-criterion class membership). An admissible system belongs to the BaseCore operational-criterion class, written $S \in \text{Crit}_{\text{op}}$, if there exists a non-empty admissible support A_S such that

$$I_{\text{per}}(S) = I_{\text{ri}}(S) = I_{\text{int}}(S) = I_{\text{cont}}(S) = I_{\text{diff}}(S) = I_{\text{leg}}(S) = 1.$$

Remark 26.6 (Neutrality of the class symbol). The symbol Crit_{op} is a shorthand for a conjunctive operational criterion. It is not a theorem of phenomenality, metaphysical subjectivity, or human equivalence.

Definition 26.7 (Rupture). A rupture is any loss of admissibility or loss of at least one critical invariant margin beyond its admissible threshold.

Theorem 26.8 (Rupture by invariant loss). *Let S be admissible. If one or more critical invariants fail beyond admissible threshold, then exactly one of the following occurs:*

1. $S \notin \text{Crit}_{\text{op}}$;
2. $\text{Part}_{\text{op}}(S)$ becomes undefined;
3. $\text{Part}_{\text{op}}(S)$ becomes trivial or non-legible in the BaseCore sense.

Proof. Failure of I_{per} destroys the admissible domain. Failure of I_{ri} destroys unique identity transport. Failure of I_{int} produces reducibility. Failure of I_{cont} destroys regime coherence. Failure of I_{diff} collapses the partition into nullity or triviality. Failure of I_{leg} leaves any remaining partition uncertifiable. These cases exhaust the six invariants. $\square$

27 Minimality of the Invariant Basis

Definition 27.1 (Reduced criterion). Let

$$\mathfrak{I} = \{I_{\text{per}}, I_{\text{ri}}, I_{\text{int}}, I_{\text{cont}}, I_{\text{diff}}, I_{\text{leg}}\}.$$

For every subset $J \subseteq \mathfrak{I}$ define the reduced class

$$\mathcal{C}_{\text{op}}(J) := \{S : I(S) = 1 \text{ for every } I \in J\}.$$

Definition 27.2 (Single-rupture witness family). Fix one invariant index $k \in \{\text{per}, \text{ri}, \text{int}, \text{cont}, \text{diff}, \text{leg}\}$. A single-rupture witness family $\mathcal{W}_k$ consists of admissible systems whose matching invariant margin vanishes while all remaining invariant margins stay strictly positive.

Proposition 27.3 (Single-rupture witnesses create false positives). *Let $J_k = \mathfrak{I} \setminus \{I_k\}$ and assume $\mathcal{W}_k \neq \emptyset$. Then every $S \in \mathcal{W}_k$ satisfies*

$$S \in \mathcal{C}_{\text{op}}(J_k) \quad \text{but} \quad S \notin \text{Crit}_{\text{op}}.$$

Proof. All retained invariants remain positive by construction, but the omitted invariant fails exactly. Proposition 25.8 then excludes S from Crit_{op} . $\square$

Theorem 27.4 (Minimality of the six-invariant basis). *Let $\mathfrak{U}$ be any admissible universe containing at least one single-rupture witness for each invariant. Then no strict subset of $\mathfrak{I}$ is extensionally equivalent to Crit_{op} on $\mathfrak{U}$.*

Proof. Take any strict subset $J \subsetneq \mathfrak{I}$. Some invariant I_k is omitted. By assumption, $\mathfrak{U}$ contains a witness $S_k \in \mathcal{W}_k$, which satisfies the reduced criterion but not Crit_{op} . Hence the two classes differ on $\mathfrak{U}$. $\square$

28 Operational Legibility as a Certified Object

Criterion 28.1 (Legibility certificate). An operational partition candidate is certified only if:

1. comparator separability holds against pre-registered baselines;
2. admissible noise does not dissolve the class map below the legibility margin;
3. the same class assignment survives on windows of length at least $\tau_{\min}(S)$;
4. targeted interventions on an invariant induce the predicted directional class change;
5. sham or off-target interventions do not generate the same class signature;
6. audit-friendly compression preserves class structure without total collapse.

Proposition 28.2 (Legibility is jointly constrained). *No single legibility metric is sufficient for I_{leg} . The invariant requires simultaneous control of separation, admissible noise robustness, persistence window, intervention fidelity, negative controls, and structured compressibility.*

Proof. A decoder can be separable yet fragile to admissible noise, stable over windows yet intervention-insensitive, or robust under noise yet collapse under compression. Because Definition 25.7 is conjunctive, all six clauses are required together. $\square$

29 Certified Candidate Audit Schema

Definition 29.1 (Base criterion certificate). A BaseCore criterion certificate for an admissible system S is a tuple

$$\text{Cert}(S) = (A_S, \mathcal{D}_S, \pi_S, \Delta(S), E_S),$$

where:

1. A_S is a certified admissible support;
2. $\mathcal{D}_S$ is a certified decoder family;
3. π_S is the decoder-induced partition on admissible histories;
4. $\Delta(S)$ is the witness margin vector;
5. E_S is the evidence bundle localizing the witness for each invariant.

Criterion 29.2 (Candidate certification rule). A candidate system may be certified as a member of Crit_{op} only if all of the following are available on the same admissible support:

1. a non-empty compact forward-invariant support certificate;
2. a positive witness margin vector $\Delta(S) > 0$ coordinate-wise;
3. a decoder family satisfying Definition 25.7;
4. a targeted intervention panel together with negative controls;
5. an auditable compression map preserving class structure;
6. a traceable dependency ledger showing which upstream result supports each invariant witness.

Theorem 29.3 (Certified criterion membership). *If $\text{Cert}(S)$ exists and satisfies Criterion 29.2, then $S \in \text{Crit}_{\text{op}}$ and $\text{Part}_{\text{op}}(S)$ is well-defined.*

Proof. Criterion 29.2 explicitly requires positive witnesses for all six invariants on a common admissible support. Definition 26.5 then gives $S \in \text{Crit}_{\text{op}}$, and Proposition 26.3 gives a well-defined partition quotient. $\square$

Remark 29.4 (Boundary discipline). The section above defines a theorem-tight grammar for certification, rupture, and falsation. It does not prove phenomenality, subjectivity, biological primacy, or cross-substrate identity transport.

Part VIII

Claim Boundary and Falsation Grammar

30 Forensic Admissibility and Integrity Grammar

Definition 30.1 (Forensic packet). Each operational run emits a packet

$$K = (\text{tick}, \text{seed}, \text{versionHash}, H_v, \text{metrics}, \text{sensorSummary}, \text{checksum}).$$

Assumption 30.2 (Packet admissibility). A run is admissible only if checksum, schema, seed consistency, version hash, and H_v consistency all pass.

Definition 30.3 (Severe violation). A run has a severe violation if any of the following holds: checksum mismatch, schema failure, seed/version inconsistency, or critical-state replay mismatch.

Criterion 30.4 (Integrity gate). Severely violated runs are excluded from support counting and marked as hard failures.

Proposition 30.5 (Integrity-first inference). *If admissibility fails, inferential significance is non-actionable.*

31 Certification, Comparison, and Rupture

Definition 31.1 (Operational certification). For an admissible system S , define

$$\text{Cert}(S) = 1 \iff I_{\text{per}}(S), I_{\text{ri}}(S), I_{\text{int}}(S), I_{\text{cont}}(S), I_{\text{diff}}(S), I_{\text{leg}}(S) > 0$$

on a non-empty admissible support under the integrity grammar above.

Definition 31.2 (Equivalence discriminator). For two admissible systems S_1, S_2 , define

$$\text{Eq}_{\Gamma, \varepsilon}(S_1, S_2) = 1$$

when the relevant signatures, admissibility bundle, decoder behavior, and invariant deltas match within the allowed comparison regime.

Definition 31.3 (Rupture event). For a targeted invariant family J , define

$$\text{Rupt}_J(S) = 1$$

when an admissible intervention or perturbation drives at least one invariant in J below its certification margin so that class membership becomes invalidated, destroyed, or undefined.

32 Prediction and Falsation Grammar

Proposition 32.1 (Complexity-only negative control prediction). *If a candidate system exhibits high connectivity, high entropy, or rich activity while failing one or more critical invariants, then*

$$\text{Cert}(S) = 0, \quad S \notin \text{Crit}_{\text{op}} \text{ in certified form,} \quad \text{Part}_{\text{op}}(S) \text{ is not certified.}$$

Proposition 32.2 (Ablation prediction). *If a system is initially certified and a targeted admissible intervention destroys one or more invariants, then the framework predicts loss of certification or transition to an ambiguity-boundary regime.*

Proposition 32.3 (Exact grammar-transport prediction). *If two realizations preserve the relevant signatures, admissibility conditions, decoder behavior, and six critical invariants up to the allowed comparison regime, then the framework predicts preserved BaseCore criterion membership.*

Proposition 32.4 (Threshold-sensitive boundary behavior). *The framework predicts stable pass/fail regions together with narrow transition bands near critical margins. It predicts neither global threshold collapse nor perfect flat insensitivity.*

Proposition 32.5 (Legibility dependence prediction). *Criterion membership becomes inadmissible if readouts cease to be separable, noise-robust, intervention-sensitive, or compressible without total collapse.*

Proposition 32.6 (Continuity and non-null relevance). *Regime continuity and non-null differentiation are constitutive. Systems that preserve rich activity while losing continuity or collapsing toward effective null structure fail the operational criterion or leave it undefined.*

Remark 32.7 (What stays outside BaseCore). Runtime status classes, Stage 2 / Stage 3 comparative programs, human comparator campaigns, Papers 7--9, and bridge-facing burden architectures remain downstream artifacts. BaseCore absorbs only the falsation grammar and claim-boundary semantics needed to keep theorem ownership honest.

33 Claim Boundary Ledger

33.1 Allowed BaseCore Claims

1. Under H1--H4, projection dynamics admit unique fixed points and compact attractor families.
2. Under H5, attractors are parameterwise non-constant.
3. Under NFD, inverse-limit identity is not determined by any finite canonical projection.
4. Under RIG-1 through RIG-5, inverse-limit structures are stable under summable admissible perturbations in the weighted metric.
5. Under NS-1 through NS-3, finite simulator classes cannot faithfully realize full CCR targets.
6. Under the six-invariant operational burden, certified criterion structures can be defined, audited, and falsified.

33.2 Forbidden BaseCore Claims

1. Human phenomenal consciousness.
2. Metaphysical subjectivity.
3. Human-machine equivalence.
4. External validation.
5. Runtime instantiation by the current system.
6. Paper 9 bridge admissibility.
7. Strong phenomenality.
8. Moral status.
9. Cross-substrate identity preservation.
10. Public claim closure.

34 Dependency Ledger

1. The foundational Hilbert-space dynamics remain the base layer.
2. Papers 1--6 contribute formal material selectively normalized into BaseCore only where theorem ownership belongs in the base.
3. Papers 7--9 remain downstream and are not dependencies of any BaseCore theorem.
4. Stage 2 / Stage 3 comparative programs remain downstream runtime and governance layers.
5. The PDF remains a derived artifact of the LaTeX tree.

35 Assumption-to-Theorem-to-Non-Claim Mapping

Assumption block	Theorem family	Non-claim boundary
H1--H4	Projection existence, contraction, unique fixed points, compact attractor family	No parameterwise non-collapse and no identity / rigidity / simulator claim
H5	Structural non-collapsibility	No claim about phenomenality, subjectivity, or runtime class membership
NFD	Non-locality of identity under canonical finite projections	No theorem that arbitrary set-theoretic injections into finite slices are impossible
RIG-1 through RIG-5	Conditional stability of inverse-limit identity and weighted Hausdorff control	No automatic isomorphism, shape equivalence, or transfer theorem without extra assumptions
NS-1 through NS-3	Conditional non-simulability and approximation barrier	No theorem that finite systems cannot approximate finite horizons at all
Six invariant burden plus legibility clauses	Criterion class membership, certified partition, rupture logic	No theorem of phenomenality, human equivalence, biology as primitive, or bridge admissibility

36 Theorem Hygiene Ledger

Theorem	Assumptions	Conclusion type	Non-conclusion	Downstream use
Existence and uniqueness of projection	H1	Hilbert-space geometric existence / uniqueness	No empirical or runtime content	Base transition grammar
Contractivity	H1--H2	Strict contraction bound	No spectrum with unit state eigenvalue	Fixed-point and spectral-gap results
Unique fixed point	H1--H3	Existence and convergence	No non-collapse by itself	Attractor family
Compactness of attractor family	H1--H4	Compactness / continuity	No regime or identity burden by itself	Parameter-family base layer
Structural non-collapsibility	H1--H5	Parameterwise exclusion of constant fixed points	No stronger semantic interpretation	Anti-collapse gate
State spectral gap	Typed differentiability at fixed point	Spectral-radius upper bound < 1	No persistence mode $\lambda = 1$ in state spectrum	Stability accounting
Non-locality under NFD	Topological regularity plus NFD	Canonical finite slices do not determine identity	No universal ban on arbitrary injections	Identity block
Conditional stability of inverse-limit identity	RIG-1 through RIG-5	Weighted Hausdorff stability under admissible perturbations	No automatic isomorphism	Rigidity accounting
Conditional non-simulability	NS-1 through NS-3	Faithful realization blocked for finite simulators	No claim against bounded finite-horizon approximation	Simulator boundary
Certified criterion membership	Six invariants plus certificate rule	Base criterion class and certified partition	No phenomenality or human-equivalence theorem	Downstream classification gates

37 Redundancy Ledger

1. The previous state-spectrum claim $\lambda_1 = 1$ under strict contraction has been removed.
2. The previous untyped Golden Mean witness has been replaced by a typed model with explicit embedding.
3. The earlier non-collapse quantifier mismatch has been removed by replacing the old H5 with a parameterwise anti-constant hypothesis.
4. The earlier absolute inverse-limit nonlocality claim has been replaced by an NFD-conditional theorem.
5. The earlier unconditional rigidity / non-simulability language has been replaced by explicit assumption blocks.
6. Substrate invariance, biological non-primacy, bridge-facing closure, and other downstream claims have been removed from BaseCore ownership.

38 Open Assumptions Inventory

1. The anti-collapse witness is abstract and theorem-valid, but not the same object as the Golden Mean typed witness.
2. Physical realization of CCR-like targets remains open.
3. Empirical instantiation of the BaseCore operational criterion remains open.
4. External validation remains open.
5. Papers 7--9 may build on BaseCore, but their additional burdens remain open at the BaseCore level.

39 Downstream Boundary Note

Papers 7, 8, and 9 remain downstream of this source tree. Their classificatory, indexed-subjectivity, and bridge-burden layers may cite BaseCore, but they are not part of the autonomous mathematical base rebuilt here.

Part IX

Discrete-to-Continuous Bridge

40 From Node Networks to Hilbert Space Dynamics

Boundary of this bridge. This part states a conditional approximation theorem between a finite node-network transition and the abstract BaseCore transition operator from Definition 5.1. It does not prove that any particular runtime execution belongs to an operational-consciousness class, and it does not convert finite tests into external validation.

Definition 40.1 (Discrete State Space). Let $N \in \mathbb{N}$ and let a node network be indexed by $i \in \{1, \dots, N\}$. Each node carries:

- a local coherence state $x_i(t) \in [0, 1]$;
- an energy coordinate $e_i(t) \in [0, 1]$;
- a phase coordinate $\phi_i(t) \in [0, 2\pi)$.

The discrete state space is

$$\mathcal{S}_N = [0, 1]^N \times [0, 1]^N \times [0, 2\pi)^N.$$

Definition 40.2 (Mean Coherence Observable). For a discrete state $s(t) = ((x_i(t))_i, (e_i(t))_i, (\phi_i(t))_i) \in \mathcal{S}_N$, define

$$X_N(t) := \frac{1}{N} \sum_{i=1}^N x_i(t) \in [0, 1].$$

This is an observable of the coherence coordinate only; it is not a complete state descriptor.

Definition 40.3 (Finite Hilbert Embedding). Let

$$\mathcal{H}_N := \mathbb{R}^N, \quad \langle \mathbf{a}, \mathbf{b} \rangle_N := \frac{1}{N} \sum_{i=1}^N a_i b_i.$$

The coherence embedding

$$\iota_N : \mathcal{S}_N \rightarrow \mathcal{H}_N$$

is

$$\iota_N((x_i)_i, (e_i)_i, (\phi_i)_i) = (x_1, \dots, x_N).$$

Thus ι_N forgets energy and phase except where they enter the transition rule below.

Definition 40.4 (Piecewise-Constant Isometry). Let $\mathcal{H} = L^2([0, 1])$ and partition $[0, 1]$ into intervals

$$I_{N,i} = \left[\frac{i-1}{N}, \frac{i}{N} \right), \quad i = 1, \dots, N.$$

Define $J_N : \mathcal{H}_N \hookrightarrow \mathcal{H}$ by

$$(J_N \mathbf{x})(s) = x_i \quad \text{for } s \in I_{N,i}.$$

Then J_N is an isometry from $(\mathcal{H}_N, \|\cdot\|_N)$ into $L^2([0, 1])$. Its adjoint $J_N^\dagger : \mathcal{H} \rightarrow \mathcal{H}_N$ is the cell-averaging map

$$(J_N^\dagger f)_i = N \int_{I_{N,i}} f(s) ds.$$

Definition 40.5 (Discrete Transition Operator). Fix auxiliary energy and phase coordinates $(e_i)_i$ and $(\phi_i)_i$, a graph with incoming neighborhoods $\text{conn}(i)$, and a control $u \in \mathcal{U}$. Let $d_N := \max_i |\text{conn}(i)|$. The coherence-coordinate transition

$$\hat{T}_{N,u}^{e,\phi} : \mathcal{H}_N \rightarrow \mathcal{H}_N$$

is defined componentwise by

$$(\hat{T}_{N,u}^{e,\phi} \mathbf{x})_i = \text{clamp}_{[0,1]}(x_i + \eta R_i(\mathbf{x}; e, \phi) + \alpha B_i(\mathbf{x}) + \Gamma_{N,i}(u)),$$

where

$$R_i(\mathbf{x}; e, \phi) = \sum_{j \in \text{conn}(i)} I_j(x_j) \frac{\exp(-|e_i - e_j|/(kT))}{\theta \cos(\phi_i - \phi_j)}.$$

Here $I_j : [0, 1] \rightarrow [0, 1]$ is the intensity readout, B_i is the finite retro-inductive feedback term, $\Gamma_N(u)$ is the finite input coupling, and $k, T, \theta, \eta, \alpha$ are runtime parameters.

Assumption 40.6 (Discrete Regularity Conditions). For a fixed parameter region, assume:

1. I_j is L_I -Lipschitz for every node j .
2. The phase denominator is uniformly separated from zero: there exists $c_\phi > 0$ such that $|\cos(\phi_i - \phi_j)| \geq c_\phi$ on every active edge.
3. $B : \mathcal{H}_N \rightarrow \mathcal{H}_N$ is L_B -Lipschitz.
4. $\Gamma_N(u)$ is independent of $\mathbf{x}$ for fixed u .
5. The parameters satisfy

$$\rho_N := \eta d_N L_I \frac{\exp(1/(kT))}{\theta c_\phi} + \alpha L_B < 1.$$

Remark 40.7 (Relation to the runtime guard). The runtime resonance function sets the contribution to zero when the denominator is numerically singular or when the Boltzmann exponent is beyond the accepted range. Assumption 40.6 is the mathematical version of that guard: without a lower bound on $|\cos(\phi_i - \phi_j)|$, the resonance quotient is not uniformly Lipschitz and no contraction theorem follows from $\theta > 1/2$ alone.

Theorem 40.8 (Discrete Operator Contractivity). *Under Assumption 40.6, for fixed u, e, ϕ the operator $\widehat{T}_{N,u}^{e,\phi}$ is a strict contraction on $\mathcal{H}_N$:*

$$\left\| \widehat{T}_{N,u}^{e,\phi}(\mathbf{x}) - \widehat{T}_{N,u}^{e,\phi}(\mathbf{y}) \right\|_N \leq \rho_N \|\mathbf{x} - \mathbf{y}\|_N \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{H}_N.$$

Proof. The clamp map on $[0, 1]^N$ is non-expansive componentwise. For each active edge, the energy-phase coefficient is bounded in absolute value by

$$\frac{\exp(1/(kT))}{\theta c_\phi}.$$

Since each I_j is L_I -Lipschitz and each row has at most d_N incoming edges, the resonance part has Lipschitz constant at most

$$d_N L_I \frac{\exp(1/(kT))}{\theta c_\phi}$$

in the normalized finite Hilbert norm. The feedback part contributes at most αL_B , and the input coupling contributes no state-dependent term. Multiplying by η and using non-expansiveness of the clamp gives the displayed bound. The assumed inequality $\rho_N < 1$ makes the contraction strict. $\square$

Assumption 40.9 (Bridge Consistency). Let $T_u = \aleph(K \cdot + \Gamma(u))$ be the BaseCore transition operator from Definition 5.1 on $\mathcal{H} = L^2([0, 1])$. Assume there are constants $C_1, C_2 > 0$ and a parameter approximation regime such that

$$\left\| J_N \widehat{T}_{N,u}^{e,\phi} J_N^\dagger - T_u \right\|_{\text{op}} \leq \varepsilon_N,$$

with

$$\varepsilon_N = \frac{C_1}{\sqrt{N}} + C_2 \max\{0, \rho_N - \|K\|\}.$$

This is a finite-rank consistency assumption, not a consequence of H1--H5 alone.

Theorem 40.10 (Hilbert--Discrete Bridge). *Assume Hypotheses 3.1--3.4, Assumptions 40.6 and 40.9, and $\rho_N < 1$. Then the embedded finite transition approximates the abstract BaseCore transition by*

$$\left\| J_N \widehat{T}_{N,u}^{e,\phi} J_N^\dagger - T_u \right\|_{\text{op}} \leq \frac{C_1}{\sqrt{N}} + C_2 \max\{0, \rho_N - \|K\|\}.$$

In particular, if $N \rightarrow \infty$, the Galerkin/quadrature term tends to zero, and the discrete parameter regime is chosen so that $\limsup_N \rho_N \leq \|K\|$, then the finite transition converges to T_u in operator norm.

Proof. The displayed estimate is exactly Assumption 40.9. The convergence statement follows because $C_1/\sqrt{N} \rightarrow 0$ and the second term vanishes under $\limsup_N \rho_N \leq \|K\|$. The theorem is therefore a bridge theorem under explicit discretization consistency, not an unconditional runtime-instantiation result. $\square$

Corollary 40.11 (Convergence of Fixed Points). *Let f_u^* be the unique fixed point of T_u from Theorem 6.1. Let $\widehat{f}_{N,u}^*$ be the unique fixed point of $\widehat{T}_{N,u}^{e,\phi}$ from Theorem 40.8. If the assumptions of Theorem 40.10 hold and*

$$q_N := \max\{\|K\|, \rho_N\} < 1,$$

then

$$\left\| J_N \widehat{f}_{N,u}^* - f_u^* \right\|_{\mathcal{H}} \leq \frac{\varepsilon_N}{1 - q_N}.$$

Consequently, if $\varepsilon_N \rightarrow 0$ and $\sup_N q_N < 1$, then

$$\lim_{N \rightarrow \infty} \left\| J_N \widehat{f}_{N,u}^* - f_u^* \right\|_{\mathcal{H}} = 0.$$

Proof. This is the standard perturbation bound for fixed points of strict contractions. Insert and subtract $T_u(J_N \hat{f}_{N,u}^*)$, use the operator discrepancy bound from Theorem 40.10, and then use the contraction constant of T_u from Theorem 5.2. Moving the contraction term to the left gives the stated estimate. $\square$

Remark 40.12 (Applicability to the Current QICN Runtime). The theorem is asymptotic and conditional. The current runtime configuration has $N_{\text{sample}} = 1500$ for heavy processing and connection count between 3 and 8, so a typical mean degree is about 5.5. The full configured potential network size is 10,000,000, but the certified finite computation uses the processing sample. Therefore the error bound is not negligible by declaration. It is a hypothesis-driven approximation bridge from finite node dynamics toward the BaseCore operator, not automatic validation of a finite execution and not evidence of phenomenality, subjecthood, or external adjudication.

A Counterexamples for Necessity

A.1 Without H1 (Convexity)

Let $\mathcal{I} = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ (unit circle, not convex).

For $\Psi = (0, 0)$, every point on $\mathcal{I}$ is equidistant. Projection is not unique.

A.2 Without H2 ($\|K\| < 1$)

Let K be a rotation by angle $\theta \neq 0$ with $\|K\| = 1$.

Orbits cycle indefinitely; no fixed point exists.

A.3 Without H3 (Completeness)

Let $\mathcal{H} = C([0, 1], \mathbb{Q})$ (rational-valued continuous functions).

Cauchy sequences may converge to irrational functions outside the space.

A.4 Without H4 (Compactness)

Let $\mathcal{U} = (0, 1)$ (open interval) and $\Gamma(u) = 1/u$.

As $u \rightarrow 0$, attractors escape to infinity; $\mathcal{A}^*$ is not compact.

A.5 Why Φ -Regularity Is a Hypothesis

Let $\mathcal{I} = \mathbb{R}$ with the Euclidean metric, let $\mathcal{E} = (-\pi/2, \pi/2)$ with the Euclidean metric, and define

$$\Phi(x) = \arctan(x).$$

The map Φ is proper as a map into the open interval $\mathcal{E}$, locally Lipschitz, and non-constant on every nonempty ball. However, no constant $C > 0$ can satisfy

$$|\arctan(x) - \arctan(y)| \geq C|x - y| \quad \forall x, y \in \mathbb{R}.$$

Indeed, with $y = x + 1$,

$$|\arctan(x + 1) - \arctan(x)| \rightarrow 0 \quad \text{as } x \rightarrow \infty,$$

while $|x - y| = 1$. Thus properness, local Lipschitz regularity, and non-collapse do not imply a global lower Lipschitz bound. Any theorem using such a bound must carry it as an explicit hypothesis or prove stronger coercive hypotheses.

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